AT&T MOBILITY

SEE AMERICAN TOWER CORPORATION MOUNT ANALYSIS REPORT DATED 03/26/2025

PROJECT:	**E/// MODERNIZATION**
SITE #:	**MPLSMNU5152**
FA #:	**10129162**
PTN #:	**3514A1EHR4, 3514A1EP5F, 3514A1EVZG, 3514A1EDY9, 3514A1EPXQ, 3514A1EFLG, 3514A1ERR4, 3514A1ESPJ, 3514A1EFL7**
IWM #:	**WSUMW0042292, WSUMW0041940, WSUMW0040434, WSUMW0040959, WSUMW0041380, WSUMW0041868 WSUMW0042327, WSUMW0040704, WSUMW0041535**
ATC ASSET#:	**311180**
JURISDICTION:	**COUNTY OF ST. LOUIS, MINNESOTA**
AT&T SITE NAME:	**LAKE VERMILLION**
ADDRESS:	**7393 FRAZIER ROAD, COOK, MN 55723-8420**

PROJECT INFORMATION

SITE NAME:	LAKE VERMILLION
COUNTY:	ST. LOUIS
ADDRESS:	7393 FRAZIER ROAD, COOK, MN 55723-8420
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ATC SITE NUMBER:	311180
LATITUDE:	47° 51' 37.565" N (47.86043476)
LONGITUDE:	92° 29' 5.225" W (-92.48478466)
TOWER OWNER:	AMERICAN TOWER CORPORATION 10 PRESIDENTIAL WAY WOBURN, MA 01801
GROUND OWNER:	KOZAR BRIAN 7393 FRAZIER ROAD COOK, MN 55723-8420
LANDLORD CONTRACT:	-
APPLICANT:	AT&T
AT&T PROJECT MANAGER:	SHAMMIKKA CHISOLM sc872r@att.com
AT&T CONSTRUCTION MANAGER:	PETER MCENERY pm753t@att.com

PROJECT CONSULTANTS

PROJECT MANAGEMENT:	AMY VALLEAU, ATC AMY.VALLEAU@AMERICANTOWER.COM
SITE ACQUISITION:	AARON ADELMAN, SMJ CONSULTING AADELMAN@SMJ-LLC.COM
ARCHITECT: (COORDINATING PROFESSIONAL)	PETER LICHOMSKI, AIA 49030 PONTIAC TRAIL, SUITE 100, WIXOM, MI 48393 PH: (248) 705-9212

SPECIAL NOTES

- ALL WORK SHALL BE INSTALLED IN CONFORMANCE WITH CURRENT AT&T CONSTRUCTION INSTALLATION GUIDE.
- EXISTING CONDITIONS WILL BE CHANGED & VERIFIED IN FIELD. IF SIGNIFICANT DEVIATIONS OR DETERIORATION ARE ENCOUNTERED AT THE TIME OF CONSTRUCTION, A REPAIR PERMIT WILL BE OBTAINED & CONTRACTOR SHALL NOTIFY ENGINEER IMMEDIATELY.
- THESE DRAWINGS ARE FULL SIZE & SCALEABLE ON 11"X17" SHEET SIZE.
- STATEMENT THAT COMPLIANCE WITH THE ENERGY CODE IS NOT REQUIRED -SCOPE OF WORK DOES NOT INVOLVE MODIFICATIONS TO EXTERIOR ENVELOPE OF BUILDING, HVAC SYSTEMS OR ELECTRICAL LIGHTING.

VICINITY MAP

LOCAL MAP

DRIVING DIRECTIONS

DIRECTIONS FROM: 7900 XERXES AVE S # 301, BLOOMINGTON, MN 55431

DEPART AND HEAD (EAST) 213 FT, TURN RIGHT 240 FT, BEAR RIGHT ONTO AMERICAN BLVD W 0.4 MI, TURN RIGHT ONTO FRANCE AVE S / COUNTY HWY-17 0.2 MI, TAKE THE SLIP ROAD ON THE RIGHT AND FOLLOW SIGNS FOR I-494 EAST 1.6 MI, AT JUNCTION 5A, HEAD RIGHT ON THE SLIP ROAD FOR I-35W NORTH TOWARDS MINNEAPOLIS 32.9 MI, TAKE THE SLIP ROAD FOR I-35 N 110.0 MI, AT JUNCTION 237, HEAD RIGHT ON THE SLIP ROAD FOR MN-33 TOWARDS CLOQUET / IRON RANGE 0.5 MI, AT THE ROUNDABOUT, TAKE THE 2ND EXIT FOR MN-33 / HIGHWAY 33 S 19.2 MI, ROAD NAME CHANGES TO US-53 N / HIGHWAY 53 47.7 MI, TAKE THE SLIP ROAD ON THE RIGHT FOR MN-169 AND HEAD TOWARDS ELY 17.6 MI, TURN LEFT ONTO ANGUS RD / COUNTY HWY-77 BP ON THE CORNER, 5.8 MI, KEEP STRAIGHT TO GET ONTO COUNTY ROAD 115 3.7 MI, TURN RIGHT ONTO FRAZER BAY RD / COUNTY HWY-418 UNPAVED ROAD, 0.3 MI, TURN LEFT 210 FT, ARRIVE AT 7393 FRAZIER ROAD, COOK, MN 55723-8420

CALL MINNESOTA ONE CALL (800) 252-1166 CALL 3 WORKING DAYS BEFORE YOU DIG!

DO NOT SCALE DRAWINGS

CONTRACTOR SHALL VERIFY ALL PLANS & EXISTING DIMENSIONS & CONDITIONS ON THE JOB SITE & SHALL IMMEDIATELY NOTIFY THE ARCHITECT OR ENGINEER IN WRITING OF ANY DISCREPANCIES BEFORE PROCEEDING WITH THE WORK OR BE RESPONSIBLE FOR SAME.

DRAWING INDEX

SCOPE OF WORKS

THIS IS NOT AN ALL INCLUSIVE LIST. CONTRACTOR SHALL UTILIZE SPECIFIED EQUIPMENT PART OR ENGINEER APPROVED EQUIVALENT. CONTRACTOR SHALL VERIFY ALL NEEDED EQUIPMENT TO PROVIDE A FUNCTIONAL SITE.
THE PROJECT GENERALLY CONSISTS OF THE FOLLOWING:

TOWER WORK:
- ROTATE (3) EXISTING AT&T ANTENNA MOUNT SECTOR FRAMES TO ACHIEVE AZIMUTH WITH NO SKEW
- REMOVE (3) EXISTING AT&T KATHREIN 80010799 PANEL ANTENNAS (1) EACH PER SECTOR IN POSITION 1 (TYP. OF 3 SECTOR)
- REMOVE (6) EXISTING AT&T ETD819G-12UB TMAS (2) EACH PER SECTOR IN POSITION 1 (TYP. OF 3 SECTOR)
- REMOVE (3) EXISTING AT&T RRH2X40-AWS RRHS (1) EACH PER SECTOR IN POSITION 2 (TYP. OF 3 SECTOR)
- REMOVE (3) EXISTING AT&T AIRSCALE RRH 4T4R B5 160W AHCA RRHS (1) EACH PER SECTOR IN POSITION 3 (TYP. OF 3 SECTOR)
- REMOVE (3) EXISTING AT&T AIRSCALE DUAL RRH 4T4R B12/B14 320W AHLBA RRHS (1) EACH PER SECTOR IN POSITION 4 (TYP. OF 3 SECTOR)
- REMOVE (3) EXISTING AT&T AHNA AIRSCALE RRH 4T4R B30 100W W/ COVER RRHS (1) EACH PER SECTOR IN POSITION 4 (TYP. OF 3 SECTOR)
- REMOVE (9) EXISTING AT&T DC2-48-60-0-9E
- REMOVE (3) EXISTING AT&T FC12-PC6-10E
- REMOVE (3) EXISTING MOUNT PIPES (1) EACH PER SECTOR IN POSITION 2 (TYP. OF 3 SECTOR)
- REMOVE (1) EXISTING 0.39" 18 PAIR FIBER TRUNK
- REMOVE (3) EXISTING 0.78" 8 AWG DC CABLES
- REMOVE (2) EXISTING 0.82" 8 AWG DC CABLES
- REMOVE (6) EXISTING 1-5/8" COAX CABLES
- RELOCATE (3) EXISTING AT&T COMMSCOPE NNH4-65C-R6 PANEL ANTENNAS (1) EACH PER SECTOR IN POSITION 4. (TYP. OF 3 SECTORS)
- REUSE (3) EXISTING AT&T COMMSCOPE NNH4-65C-R6 ANTENNAS (1) EACH PER SECTOR IN POSITION 3 . (TYP. OF 3 SECTORS)
- REUSE (6) EXISTING 1-5/8" COAX CABLES
- REUSE (1) EXISTING 3/8" RET CONTROL CABLE
- INSTALL (3) PROPOSED AT&T ERICSSON AIR6472 B77G B77M PANEL ANTENNAS (1) EACH PER SECTOR IN POSITION 2 (TYP. OF 3 SECTORS)
- INSTALL (3) PROPOSED AT&T RRHS 4490 B5/B12A (1) EACH PER SECTOR IN POSITION 1. (TYP. OF 3 SECTORS)
- INSTALL (3) PROPOSED AT&T RRHS 4471 B30 (1) EACH PER SECTOR IN POSITION 1. (TYP. OF 3 SECTORS)
- INSTALL (3) PROPOSED AT&T RRHS 4494 B14/B29 (1) EACH PER SECTOR IN POSITION 3. (TYP. OF 3 SECTORS)
- INSTALL (3) PROPOSED AT&T RRHS 4890 B25/B66 (1) EACH PER SECTOR IN POSITION 3. (TYP. OF 3 SECTORS)
- INSTALL (2) PROPOSED DC9-48-60-24-8C-EV (ENCLOSURE)
- INSTALL (2) PROPOSED 0.40" 24 PAIR FIBER TRUNKS
- INSTALL (5) PROPOSED 0.96" 6 AWG DC CABLES
- INSTALL (6) D200RRU B2B OR APPROVED EQUAL RRH MOUNTS
- INSTALL (3) SITE PRO 1 - SFS-H-L VALMONT V-STABILIZER (INSTALL HORIZONTALLY LEVEL) (FIELD CUT TO FINAL LENGTH 59").
- INSTALL (9) SITE PRO 1 - HMB-AU VALMONT HANDRAIL ANTENNA PIPE MOUNTING BRACKET KIT
- INSTALL (3) (2.875"OD X 0.203" X 120") IN MOUNTING PIPE POSITION 2
- INSTALL (3) BULK PIPE (2.875"OD X 0.203" X 150")
- INSTALL (15) SITE PRO 1 - SCX7-U VALMONT CROSSOVER KIT
- INSTALL (3) SITE PRO 1 - SPTB VALMONT TIE-BACK ASSEMBLY
- INSTALL (6) BULK PIPE (2.375"OD X 0.154" X 72")
- INSTALL AT&T APPROVED HOISTING GRIPS (COMMSCOPE PART#: 860674608-U)

GROUND WORK:

POWER PLANT:
- REMOVE (5) C48/24-1500 CONVERTERS (NO RETAINED CIRCUITS)
- REWIRE FIF FUSE PANEL FROM +24V TO -48V
- REUSE (1) VERTIV STD -48VDC NETSURE 7100 PLANT 1000A
- REUSE (14) R48-2000E3 RECTIFIERS - RELOCATE FROM CONVERTER SLOTS TO RECITIFER SLOTS AS NEEDED
- REUSE (20) POWERSAFE SBS190F BATTERIES (INSTALL DATE: 10/13/2020)
- REUSE (8) 200A BATTERY DISCONNECT BREAKERS
- INSTALL (1) -58V UPGRADE KIT
- INSTALL (9) -58V CONVERTERS
- INSTALL BREAKERS AS NEEDED PER ATT-CEM-18002

CIVIL:
- REMOVE NOKIA BBU EQUIPMENT AFTER SITE CUTOVER
- REMOVE COMMSCOPE POWERSHFT AND REWIRE CIRCUITS TO POWERPLANT
- REUSE (1) 7705 SIAD
- REUSE (1) GENERIC NID
- REUSE (1) DC12-48-60-RM (GEN 1)
- REUSE (1) DC12-48-60-RM (GEN 2)
- INSTALL (1) 6672 BBU CARD, (1) 6610 SITE CONTROLLER
- INSTALL E/// BBU EQUIPMENT IN (INSTALL IN EXISTING FIF RACK)

CODE COMPLIANCE

- 2023 NATIONAL ELECTRICAL CODE (NEC 2023) EFFECTIVE 7/1/23
- 2020 MN BUILDING CODE - MINNESOTA (2018 IBC W/ AMENDMENT)
- 2020 MN MECHANICAL CODE - MINNESOTA
- ANSI/TIA-222-H

REFERENCE MATERIALS

- THESE DRAWINGS ARE BASED AT&T SCOPING DOCUMENT DATED 10/16/2024
- REVISED RFDS PENDING. CONTRACTOR TO USE LATEST VERSION WITH CD'S PER SCOPE OF WORK.

AT&T
7900 XERXES AVE S, 3RD FLOOR BLOOMINGTON, MN 55431

AMERICAN TOWER
SITE DESIGN
3500 REGENCY PARKWAY, STE 100 CARY, NORTH CAROLINA 27518 PHONE: (919) 468-0112

LAB
49030 Pontiac Trail, Suite 100
Wixom, Michigan 48393
PHONE: (248) 705-9212

REVISIONS

REV.	DATE	DESCRIPTION	INITIALS
A	04/07/25	ISSUED FOR REVIEW	RC
0	04/17/2025	FINAL CDS	RC

NOT FOR CONSTRUCTION UNLESS LABELED AS CONSTRUCTION SET

I HEREBY CERTIFY THAT THIS PLAN, SPECIFICATION, OR REPORT WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED ARCHITECT UNDER THE LAWS OF THE STATE OF MINNESOTA.

SIGNATURE:

PRINTED NAME: PETER LICHOMSKI
DATE: 04/17/2025
LICENSE NUMBER: 46120

PETER LICHOMSKI
LICENSED ARCHITECT
No. 46120
STATE OF MINNESOTA

E/// MODERNIZATION
10129162
LAKE VERMILLION
7393 FRAZIER ROAD,
COOK, MN 55723-8420

SHEET TITLE

TITLE SHEET

SHEET NUMBER

T1

GENERAL CONSTRUCTION

1. FOR THE PURPOSE OF CONSTRUCTION DRAWINGS, THE FOLLOWING DEFINITIONS SHALL APPLY:
   CONTRACTOR/CM -MASTEC
   SUB-CONTRACTOR - PER TRADE
   OWNER- AT&T WIRELESS
2. ALL SITE WORK SHALL BE COMPLETED AS INDICATED ON THE DRAWINGS AND AT&T PROJECT SPECIFICATIONS.
3. GENERAL CONTRACTOR SHALL VISIT THE SITE AND SHALL FAMILIARIZE HIMSELF WITH ALL CONDITIONS AFFECTING THE PROPOSED WORK AND SHALL MAKE PROVISIONS. GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR FAMILIARIZING HIMSELF WITH ALL CONTRACT DOCUMENTS, FIELD CONDITIONS, DIMENSIONS, AND CONFIRMING THAT THE WORK MAY BE ACCOMPLISHED AS SHOWN PRIOR TO PROCEEDING WITH CONSTRUCTION. ANY DISCREPANCIES SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER PRIOR TO THE COMMENCEMENT OF WORK.
4. ALL MATERIALS FURNISHED AND INSTALLED SHALL BE IN STRICT ACCORDANCE WITH ALL APPLICABLE CODES, REGULATIONS, AND ORDINANCES. GENERAL CONTRACTOR SHALL ISSUE ALL APPROPRIATE NOTICES AND COMPLY WITH ALL LAWS, ORDINANCES, RULES, REGULATIONS, AND LAWFUL ORDERS OF ANY PUBLIC AUTHORITY REGARDING THE PERFORMANCE OF WORK.
5. ALL WORK CARRIED OUT SHALL COMPLY WITH ALL APPLICABLE MUNICIPAL AND UTILITY COMPANY SPECIFICATIONS AND LOCAL JURISDICTIONAL CODES, ORDINANCES, AND APPLICABLE REGULATIONS.
6. UNLESS NOTED OTHERWISE, THE WORK SHALL INCLUDE FURNISHING MATERIALS, EQUIPMENT, APPURTENANCES, AND LABOR NECESSARY TO COMPLETE ALL INSTALLATIONS AS INDICATED ON THE DRAWINGS.
7. PLANS ARE NOT TO BE SCALED. THESE PLANS ARE INTENDED TO BE A DIAGRAMMATIC OUTLINE ONLY UNLESS OTHERWISE NOTED. DIMENSIONS SHOWN ARE TO FINISH SURFACES UNLESS OTHERWISE NOTED. SPACING BETWEEN EQUIPMENT IS THE MINIMUM REQUIRED CLEARANCE. THEREFORE, IT IS CRITICAL TO FIELD VERIFY DIMENSIONS, SHOULD THERE BE ANY QUESTIONS REGARDING THE CONTRACT DOCUMENTS, THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING A CLARIFICATION FROM THE ENGINEER PRIOR TO PROCEEDING WITH THE WORK. DETAILS ARE INTENDED TO SHOW DESIGN INTENT. MODIFICATIONS MAY BE REQUIRED TO SUIT JOB DIMENSIONS OR CONDITIONS AND SUCH MODIFICATIONS SHALL BE INCLUDED AS PART OF WORK AND PREPARED BY THE ENGINEER PRIOR TO PROCEEDING WITH WORK.
8. THE CONTRACTOR SHALL INSTALL ALL EQUIPMENT AND MATERIALS IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS UNLESS SPECIFICALLY STATED OTHERWISE.
9. IF THE SPECIFIED EQUIPMENT CANNOT BE INSTALLED AS SHOWN ON THESE DRAWINGS, THE CONTRACTOR SHALL PROPOSE AN ALTERNATIVE INSTALLATION SPACE FOR APPROVAL BY THE ENGINEER PRIOR TO PROCEEDING.
10. GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR THE SAFETY OF WORK AREA, ADJACENT AREAS AND BUILDING OCCUPANTS THAT ARE LIKELY TO BE AFFECTED BY THE WORK UNDER THIS CONTRACT. WORK SHALL CONFIRM TO ALL OSHA REQUIREMENTS AND THE LOCAL JURISDICTION.
11. GENERAL CONTRACTOR SHALL COORDINATE WORK AND SCHEDULE WORK ACTIVITIES WITH OTHER DISCIPLINES.
12. ERECTION SHALL BE DONE IN A WORKMANLIKE MANNER BY COMPETENT EXPERIENCED WORKMAN IN ACCORDANCE WITH APPLICABLE CODES AND THE BEST ACCEPTED PRACTICE. ALL MEMBERS SHALL BE LAID PLUMB AND TRUE AS INDICATED ON THE DRAWINGS.
13. SEAL PENETRATIONS THROUGH FIRE RATED AREAS WITH UL LISTED MATERIALS APPROVED BY LOCAL JURISDICTION. CONTRACTOR SHALL KEEP AREA CLEAN, HAZARD FREE, AND DISPOSE OF ALL DEBRIS.
14. WORK PREVIOUSLY COMPLETED IS REPRESENTED BY LIGHT SHADED LINES AND NOTES. THE SCOPE OF WORK FOR THIS PROJECT IS REPRESENTED BY DARK SHADED LINES AND NOTES. CONTRACTOR SHALL NOTIFY THE GENERAL CONTRACTOR OF ANY EXISTING CONDITIONS THAT DEVIATE FROM THE DRAWINGS PRIOR TO BEGINNING CONSTRUCTION.
15. CONTRACTOR SHALL PROVIDE WRITTEN NOTICE TO THE CONSTRUCTION MANAGER 48 HOURS PRIOR TO COMMENCEMENT OF WORK.
16. THE CONTRACTOR SHALL PROTECT EXISTING IMPROVEMENTS, PAVEMENTS, CURBS, LANDSCAPING AND STRUCTURES. ANY DAMAGED PART SHALL BE REPAIRED AT CONTRACTOR'S EXPENSE TO THE SATISFACTION OF THE OWNER.
17. THE CONTRACTOR SHALL CONTACT UTILITY LOCATING SERVICES PRIOR TO THE START OF CONSTRUCTION.
18. GENERAL CONTRACTOR SHALL COORDINATE AND MAINTAIN ACCESS FOR ALL TRADES AND CONTRACTORS TO THE SITE AND/OR BUILDING.
19. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR SECURITY OF THE SITE FOR THE DURATION OF CONSTRUCTION UNTIL JOB COMPLETION.
20. THE GENERAL CONTRACTOR SHALL MAINTAIN IN GOOD CONDITION ONE COMPLETE SET OF PLANS WITH ALL REVISIONS, ADDENDA, AND CHANGE ORDERS ON THE PREMISES AT ALL TIMES.
21. THE GENERAL CONTRACTOR SHALL PROVIDE PORTABLE FIRE EXTINGUISHERS WITH A RATING OF NOT LESS THAN 2-A OT 2-A:10-B:C AND SHALL BE WITHIN 25 FEET OF TRAVEL DISTANCE TO ALL PORTIONS OF WHERE THE WORK IS BEING COMPLETED DURING CONSTRUCTION.
22. ALL EXISTING ACTIVE SEWER, WATER, GAS, ELECTRIC, AND OTHER UTILITIES SHALL BE PROTECTED AT ALL TIMES, AND WHERE REQUIRED FOR THE PROPER EXECUTION OF THE WORK, SHALL BE RELOCATED AS DIRECTED BY THE ENGINEER. EXTREME CAUTION SHOULD BE USED BY THE CONTRACTOR WHEN EXCAVATING OR DRILLING PIERS AROUND OR NEAR UTILITIES. CONTRACTOR SHALL PROVIDE SAFETY TRAINING FOR THE WORKING CREW. THIS SHALL INCLUDE BUT NOT BE LIMITED TO A) FALL PROTECTION, B) CONFINED SPACE, C) ELECTRICAL SAFETY, AND D) TRENCHING & EXCAVATION.
23. ALL EXISTING INACTIVE SEWER, WATER, GAS, ELECTRIC, AND OTHER UTILITIES, WHICH INTERFERE WITH THE EXECUTION OF THE WORK, SHALL BE REMOVED, CAPPED, PLUGGED OR OTHERWISE DISCONNECTED AT POINTS WHICH WILL NOT INTERFERE WITH THE EXECUTION OF THE WORK, AS DIRECTED BY THE RESPONSIBLE ENGINEER, AND SUBJECT TO THE APPROVAL OF THE OWNER AND/OR LOCAL UTILITIES.
24. THE AREAS OF THE OWNER'S PROPERTY DISTURBED BY THE WORK AND NOT COVERED BY THE TOWER, EQUIPMENT OR DRIVEWAY, SHALL BE GRADED TO A UNIFORM SLOPE AND STABILIZED TO PREVENT EROSION.
25. CONTRACTOR SHALL MINIMIZE DISTURBANCE TO THE EXISTING SITE DURING CONSTRUCTION. EROSION CONTROL MEASURES, IF REQUIRED DURING CONSTRUCTION, SHALL BE IN CONFORMANCE WITH THE FEDERAL AND LOCAL JURISDICTION FOR EROSION AND SEDIMENT CONTROL.
26. NO FILL OR EMBANKMENT MATERIAL SHALL BE PLACED ON FROZEN GROUNDING, FROZEN MATERIALS, SNOW OR ICE SHALL NOT BE PLACED IN ANY FILL OR EMBANKMENT.
27. THE SUBGRADE SHALL BE BROUGHT TO A SMOOTH UNIFORM GRADE AND COMPACTED TO 95 PERCENT STANDARD PROCTOR DENSITY UNDER PAVEMENT AND STRUCTURES AND 80 PERCENT STANDARD PROCTOR DENSITY IN OPEN SPACE. ALL TRENCHES IN PUBLIC RIGHT OF WAY SHALL BE BACKFILLED WITH FLOWABLE FILL OR OTHER MATERIAL PRE-APPROVED BY THE LOCAL JURISDICTION.
28. ALL NECESSARY RUBBISH, STUMPS, DEBRIS, STICKS, STONES, AND OTHER REFUSE SHALL BE REMOVED FROM THE SITE AND DISPOSED OF IN A LAWFUL MANNER.
29. ALL BROCHURES, OPERATING AND MAINTENANCE MANUALS, CATALOGS, SHOP DRAWINGS AND OTHER DOCUMENTS SHALL BE TURNED OVER TO THE GENERAL CONTRACTOR AT COMPLETION OF CONSTRUCTION AND PRIOR TO PAYMENT.
30. CONTRACTOR SHALL SUBMIT A COMPLETE SET OF AS-BUILT REDLINES TO THE GENERAL CONTRACTOR UPON COMPLETION OF PROJECT AND PRIOR TO FINAL PAYMENT.
31. CONTRACTOR SHALL LEAVE PREMISES IN A CLEAN CONDITION.
32. THE PROPOSED FACILITY WILL BE UNMANNED AND DOES NOT REQUIRE POTABLE WATER OR SEWER SERVICE, AND IS NOT FOR HUMAN HABITAT (NO HANDICAP ACCESS REQUIRED).
33. OCCUPANCY IS LIMITED TO PERIODIC MAINTENANCE AND INSPECTION, APPROXIMATELY 2 TIMES PER MONTH, BY AT&T TECHNICIANS.
34. NO OUTDOOR STORAGE OR SOLID WASTE CONTAINERS ARE PROPOSED.
35. ALL MATERIAL SHALL BE FURNISHED AND WORK SHALL BE PERFORMED IN ACCORDANCE WITH THE LATEST REVISION AT&T MOBILITY GROUNDING STANDARD "TECHNICAL SPECIFICATION FOR CONSTRUCTION OF GSM/GPRS WIRELESS SITES" AND "TECHNICAL SPECIFICATION FOR FACILITY GROUNDING". IN CASE OF A CONFLICT BETWEEN THE CONSTRUCTION SPECIFICATION AND THE DRAWINGS, THE DRAWINGS SHALL GOVERN.
36. CONTRACTORS SHALL BE RESPONSIBLE FOR OBTAINING ALL PERMITS AND INSPECTIONS REQUIRED FOR CONSTRUCTION. IF CONTRACTOR CANNOT OBTAIN A PERMIT, THEY MUST NOTIFY THE GENERAL CONTRACTOR IMMEDIATELY.
37. CONTRACTOR SHALL REMOVE ALL TRASH AND DEBRIS FROM THE SITE ON A DAILY BASIS.
38. INFORMATION SHOWN ON THESE DRAWINGS WAS OBTAINED FROM SITE VISITS AND/OR DRAWINGS PROVIDED BY THE SITE OWNER. CONTRACTORS SHALL NOTIFY THE ENGINEER OF ANY DISCREPANCIES PRIOR TO ORDERING MATERIAL OR PROCEEDING WITH CONSTRUCTION.
39. NO WHITE STROBE LIGHTS ARE PERMITTED LIGHTING IF REQUIRED, WILL MEET FAA STANDARDS AND REQUIREMENTS.

ANTENNA MOUNTING

40. DESIGN AND CONSTRUCTION OF ANTENNA SUPPORTS SHALL CONFORM TO CURRENT ANSI/TIA-222 OR APPLICABLE LOCAL CODES.
41. ALL STEEL MATERIALS SHALL BE GALVANIZED AFTER FABRICATION IN ACCORDANCE WITH ASTM A123 "ZINC (HOT-DIP GALVANIZED) COATINGS ON IRON AND STEEL PRODUCTS", UNLESS NOTED OTHERWISE.
42. ALL BOLTS, ANCHORS AND MISCELLANEOUS HARDWARE SHALL BE GALVANIZED IN ACCORDANCE WITH ASTM A153 "ZINC-COATING (HOT-DIP) ON IRON AND STEEL HARDWARE", UNLESS NOTED OTHERWISE.
43. DAMAGED GALVANIZED SURFACES SHALL BE REPAIRED BY COLD GALVANIZING IN ACCORDANCE WITH ASTM A780.
44. ALL ANTENNA MOUNTS SHALL BE INSTALLED WITH LOCK NUTS, DOUBLE NUTS AND SHALL BE TORQUED TO MANUFACTURER'S RECOMMENDATIONS.
45. CONTRACTOR SHALL INSTALL ANTENNA PER MANUFACTURER'S RECOMMENDATION FOR INSTALLATION AND GROUNDING.
46. ALL UNUSED PORTS ON ANY ANTENNAS SHALL BE TERMINATED WITH A 50-OHM LOAD TO ENSURE ANTENNAS PERFORM AS DESIGNED.
47. PRIOR TO SETTING ANTENNA AZIMUTHS AND DOWNTILTS, ANTENNA CONTRACTOR SHALL CHECK THE ANTENNA MOUNT FOR TIGHTNESS AND ENSURE THAT THEY ARE PLUMB. ANTENNA AZIMUTHS SHALL BE SET FROM TRUE NORTH AND BE ORIENTED WITHIN +/-5% AS DEFINED BY THE RFDS ANTENNA DOWNTILTS SHALL BE WITHIN +/- 0 5% AS DEFINED BY THE RFDS REFER TO ND-00246.
48. JUMPERS FROM THE TMA'S MUST TERMINATE TO OPPOSITE POLARIZATION'S IN EACH SECTOR.
49. CONTRACTOR SHALL RECORD THE SERIAL #, SECTOR AND POSITION OF EACH ACTUATOR INSTALLED AT THE ANTENNAS AND PROVIDE THE INFORMATION TO AT&T.
50. TMA'S SHALL BE MOUNTED ON PIPE DIRECTLY BEHIND ANTENNAS AS CLOSE TO ANTENNA AS FEASIBLE IN A VERTICAL POSITION.

TORQUE REQUIREMENTS

51. ALL RF CONNECTIONS SHALL BE TIGHTENED BY A TORQUE WRENCH.
52. ALL RF CONNECTIONS, GROUNDING HARDWARE AND ANTENNA HARDWARE SHALL HAVE A TORQUE MARK INSTALLED IN A CONTINUOUS STRAIGHT LINE FROM BOTH SIDES OF THE CONNECTION.
   A. RF CONNECTION BOTH SIDES OF THE CONNECTOR.
   B. GROUNDING AND ANTENNA HARDWARE ON THE NUT SIDE STARTING FROM THE THREADS TO THE SOLID SURFACE. EXAMPLE OF SOLID SURFACE: GROUND BAR, ANTENNA BRACKET METAL.

FIBER & POWER CABLE MOUNTING

53. THE FIBER OPTIC TRUNK CABLES SHALL BE INSTALLED INTO CONDUITS, CHANNEL CABLE TRAYS, OR CABLE TRAY. WHEN INSTALLING FIBER OPTIC TRUNK CABLES INTO A CABLE TRAY SYSTEM, THEY SHALL BE INSTALLED INTO AN INTER DUCT AND A PARTITION BARRIER SHALL BE INSTALLED BETWEEN THE 600 VOLT CABLES AND THE INTER DUCT IN ORDER TO SEGREGATE CABLE TYPES. OPTIC FIBER TRUNK CABLES SHALL HAVE APPROVED CABLE RESTRAINTS EVERY (60) SIXTY FEET AND SECURELY FASTENED TO THE CABLE TRAY SYSTEM. NFPA 70 (NEC) ARTICLE 770 RULES SHALL APPLY.
54. THE TYPE TC-ER CABLES SHALL BE INSTALLED INTO CONDUITS, CHANNEL CABLE TRAYS, OR CABLE TRAY AND SHALL BE SECURED AT INTERVALS NOT EXCEEDING (6) SIX FEET. AN EXCEPTION; WHERE TYPE TC-ER CABLES ARE NOT SUBJECT TO PHYSICAL DAMAGE, CABLES SHALL BE PERMITTED TO MAKE A TRANSITION BETWEEN CONDUITS, CHANNEL CABLE TRAYS, OR CABLE TRAY WHICH ARE SERVING UTILIZATION EQUIPMENT OR DEVICES, A DISTANCE (6) SIX FEET SHALL NOT BE EXCEEDED WITHOUT CONTINUOUS SUPPORTING. NFPA 70 (NEC) ARTICLES 336 AND 392 RULES SHALL APPLY.
55. WHEN INSTALLING OPTIC FIBER TRUNK CABLES OR TYPE TC-ER CABLES INTO CONDUITS, NFPA 70 (NEC) ARTICLE 300 RULES SHALL APPLY.

56-61 RESERVED FOR ADDITIONAL NOTES.

COAXIAL CABLE NOTES

62. TYPES AND SIZES OF THE ANTENNA CABLE ARE BASED ON ESTIMATED LENGTHS. PRIOR TO ORDERING CABLE, CONTRACTOR SHALL VERIFY ACTUAL LENGTH BASED ON CONSTRUCTION LAYOUT AND NOTIFY THE PROJECT MANAGER IF ACTUAL LENGTHS EXCEED ESTIMATED LENGTHS.
63. CONTRACTOR SHALL VERIFY THE DOWN-TILT OF EACH ANTENNA WITH A DIGITAL LEVEL.
64. CONTRACTOR SHALL CONFIRM COAX COLOR CODING PRIOR TO CONSTRUCTION.
65. ALL JUMPERS TO THE ANTENNAS FROM THE MAIN TRANSMISSION LINE SHALL BE 1/2" DIA LDF AND SHALL NOT EXCEED 6'-0".
66. ALL COAXIAL CABLE SHALL BE SECURED TO THE DESIGNED SUPPORT STRUCTURE, IN AN APPROVED MANNER, AT DISTANCES NOT TO EXCEED 4'-0" OC.
67. CONTRACTOR SHALL FOLLOW ALL MANUFACTURER'S RECOMMENDATIONS REGARDING BOTH THE INSTALLATION AND GROUNDING OF ALL COAXIAL CABLES, CONNECTORS, ANTENNAS, AND ALL OTHER EQUIPMENT.
68. CONTRACTOR SHALL GROUND ALL EQUIPMENT INCLUDING ANTENNAS, RET MOTORS, TMA'S, COAX CABLES, AND RET CONTROL CABLES AS A COMPLETE SYSTEM. GROUNDING SHALL BE EXECUTED BY QUALIFIED WIREMEN IN COMPLIANCE WITH MANUFACTURER'S SPECIFICATION AND RECOMMENDATION.
69. CONTRACTOR SHALL PROVIDE STRAIN-RELIEF AND CABLE SUPPORTS FOR ALL CABLE ASSEMBLIES, COAX CABLES, AND RET CONTROL CABLES. CABLE STRAIN-RELIEFS AND CABLE SUPPORTS SHALL BE APPROVED FOR THE PURPOSE. INSTALLATION SHALL BE IN ACCORDANCE WITH MANUFACTURER'S SPECIFICATIONS AND RECOMMENDATIONS.
70. CONTRACTOR TO VERIFY THAT EXISTING COAX HANGERS ARE STACKABLE SNAP IN HANGERS. IF EXISTING HANGERS ARE NOT STACKABLE SNAP IN HANGERS THE CONTRACTOR SHALL REPLACE EXISTING HANGERS WITH NEW SNAP IN HANGERS IF APPLICABLE.

GENERAL CABLE AND EQUIPMENT NOTES

71. CONTRACTOR SHALL BE RESPONSIBLE TO VERIFY ANTENNA, TMAS, DIPLEXERS, AND COAX CONFIGURATION, MAKE AND MODELS PRIOR TO INSTALLATION.
72. ALL CONNECTIONS FOR HANGERS, SUPPORTS, BRACING, ETC SHALL BE INSTALLED PER TOWER MANUFACTURER'S RECOMMENDATIONS.
73. CONTRACTOR SHALL REFERENCE THE TOWER STRUCTURAL ANALYSIS/DESIGN DRAWINGS FOR DIRECTIONS ON CABLE DISTRIBUTION/ROUTING.
74. ALL OUTDOOR RF CONNECTORS/CONNECTIONS SHALL BE WEATHERPROOFED, EXCEPT THE RET CONNECTORS, USING BUTYL TAPE AFTER INSTALLATION AND FINAL CONNECTIONS ARE MADE. BUTYL TAPE SHALL HAVE A MINIMUM OF ONE-HALF TAPE WIDTH OVERLAP ON EACH TURN AND EACH LAYER SHALL BE WRAPPED THREE TIMES. WEATHERPROOFING SHALL BE SMOOTH WITHOUT BUCKLING BUTYL BLEEDING IS NOT ALLOWED.
75. IF REQUIRED TO PAINT ANTENNAS AND/OR COAX:
   A. TEMPERATURE SHALL BE ABOVE 50° F.
   B. PAINT COLOR MUST BE APPROVED BY BUILDING OWNER/LANDLORD.
   C. FOR REGULATED TOWERS, FAA/FCC APPROVED PAINT IS REQUIRED.
   D. DO NOT PAINT OVER COLOR CODING OR ON EQUIPMENT MODEL NUMBERS.
76. ALL CABLES SHALL BE GROUNDED WITH COAXIAL CABLE GROUND KITS FOLLOW THE MANUFACTURER'S RECOMMENDATIONS.
   A. GROUNDING AT THE ANTENNA LEVEL.
   B. GROUNDING AT MID LEVEL, TOWERS WHICH ARE OVER 200'-0", ADDITIONAL CABLE GROUNDING REQUIRED.
   C. GROUNDING AT BASE OF TOWER PRIOR TO TURNING HORIZONTAL.
   D. GROUNDING OUTSIDE THE EQUIPMENT SHELTER AT ENTRY PORT.
   E. GROUNDING INSIDE THE EQUIPMENT SHELTER AT THE ENTRY PORT.
77. ALL PROPOSED GROUND BAR DOWNLEADS ARE TO BE TERMINATED TO THE EXISTING ADJACENT GROUND BAR DOWNLEADS A MINIMUM DISTANCE OF 4'-0" BELOW GROUND BAR. TERMINATIONS MAY BE EXOTHERMIC OR COMPRESSION.

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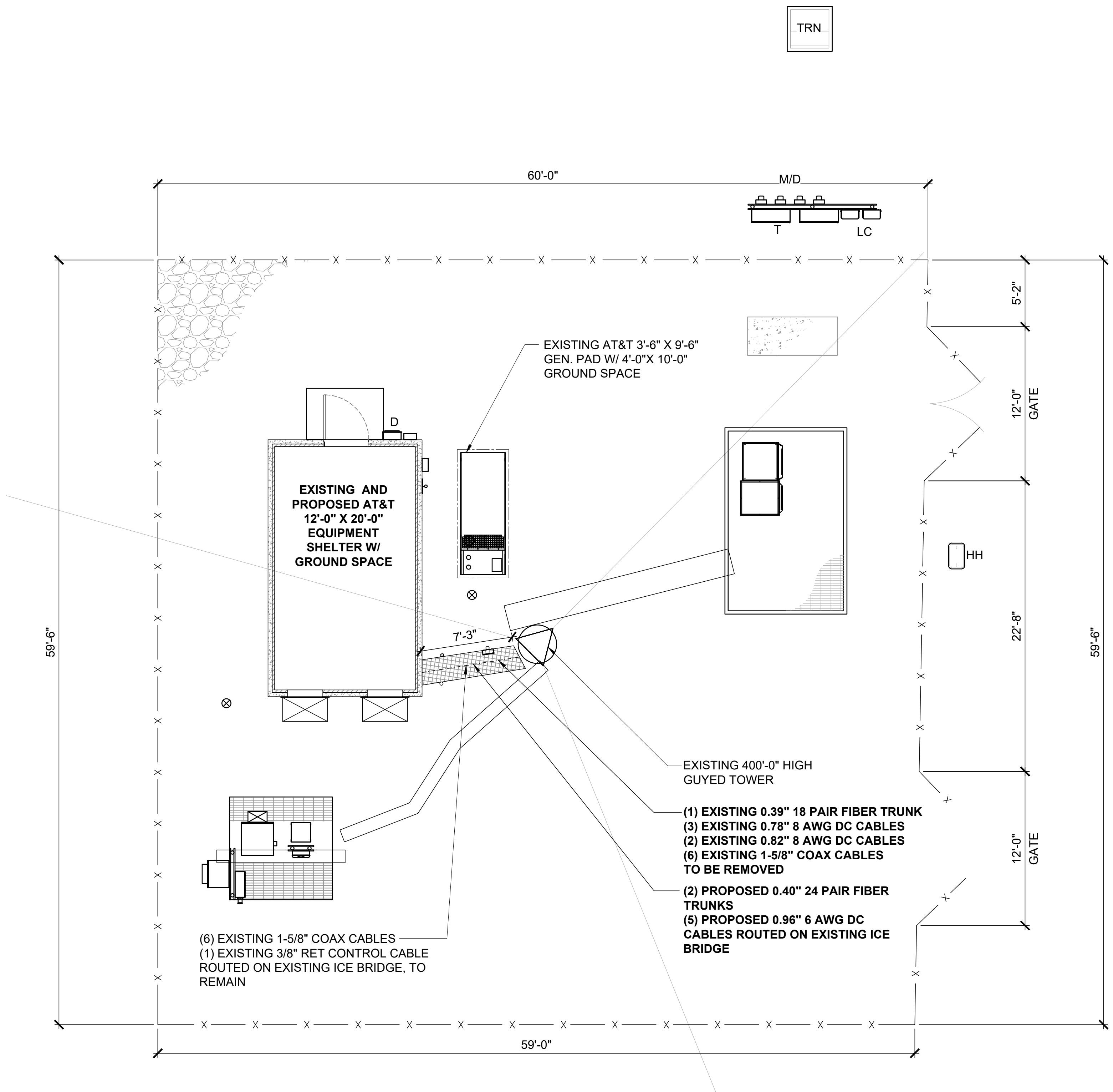
COOK, MN 55723-8420

SHEET TITLE

NOTES & SPECIFICATIONS

SHEET NUMBER

SP1



NORTH

COMPOUND PLAN

22"x34" SCALE: 3/16" = 1'-0"
11"x17" SCALE: 3/32" = 1'-0"

1

SITE PHOTO 1

3

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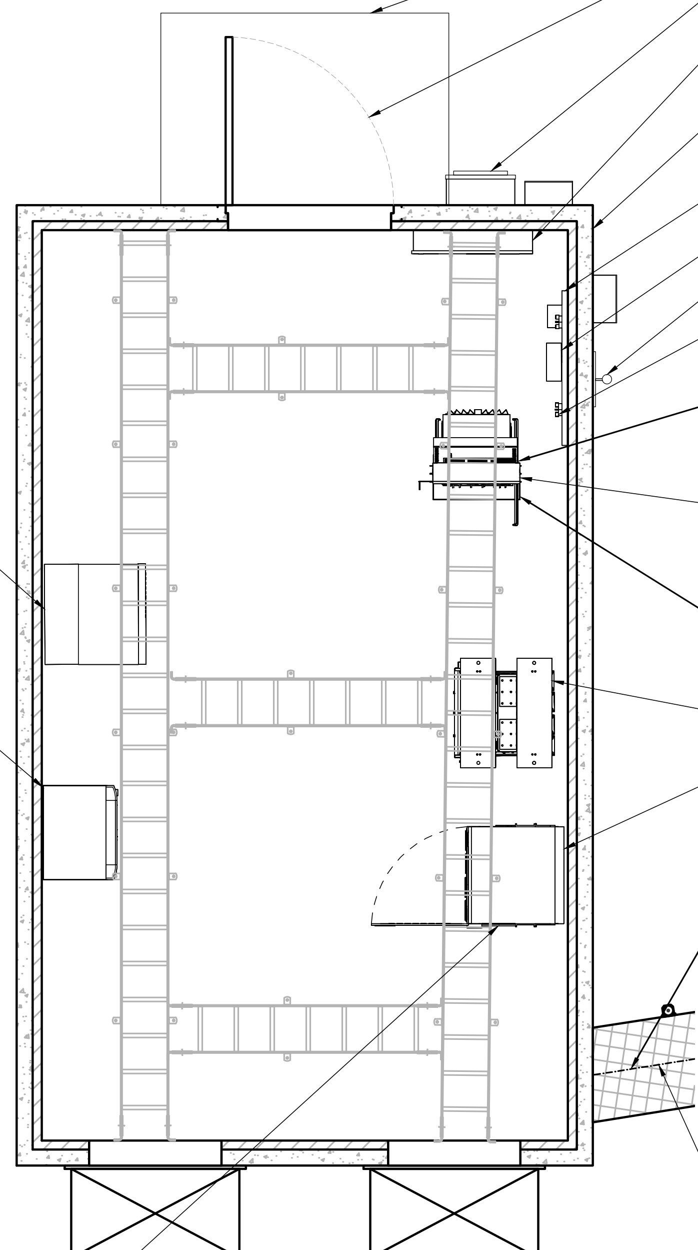
COMPOUND PLAN

SHEET NUMBER

A1

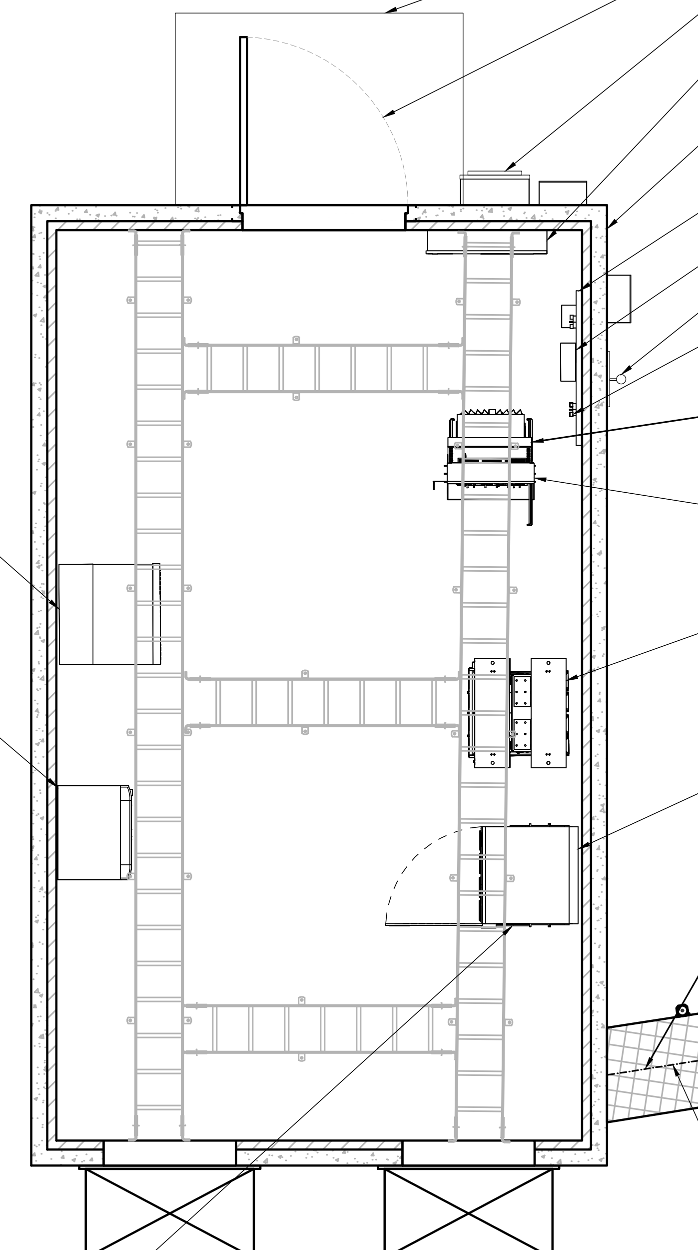


POWER PLANT:

EXISTING AT&T (1) VERTIV STD -48VDC NETSURE 7100 PLANT 1000A
EXISTING (14) R48-2000E3 RECTIFIERS - RELOCATE FROM CONVERTER SLOTS TO RECITIFER SLOTS AS NEEDED, EXISTING (20) POWERSAFE SBS190F BATTERIES (INSTALL DATE: 10/13/2020), EXISTING (8) 200A BATTERY DISCONNECT BREAKERS
EXISTING POWER PLANT (TO REMAIN)
EXISTING AT&T (5) C48/24-1500 CONVERTERS (NO RETAINED CIRCUITS) (TO BE REMOVED)

EXISTING EQUIPMENT PLAN | 22"x34" SCALE: 1/2" = 1'-0" | 11"x17" SCALE: 1/4" = 1'-0" | 1

INSTALL BREAKERS AS NEEDED PER ATT-CEM-18002



POWER PLANT:

EXISTING AT&T (1) VERTIV STD -48VDC NETSURE 7100 PLANT 1000A
EXISTING (14) R48-2000E3 RECTIFIERS - RELOCATE FROM CONVERTER SLOTS TO RECITIFER SLOTS AS NEEDED, EXISTING (20) POWERSAFE SBS190F BATTERIES (INSTALL DATE: 10/13/2020), EXISTING (8) 200A BATTERY DISCONNECT BREAKERS
EXISTING POWER PLANT (TO REMAIN)
INSTALL (9) -58V CONVERTERS
INSTALL (1) -58V UPGRADE KIT

PROPOSED EQUIPMENT PLAN | 22"x34" SCALE: 1/2" = 1'-0" | 11"x17" SCALE: 1/4" = 1'-0" | 2

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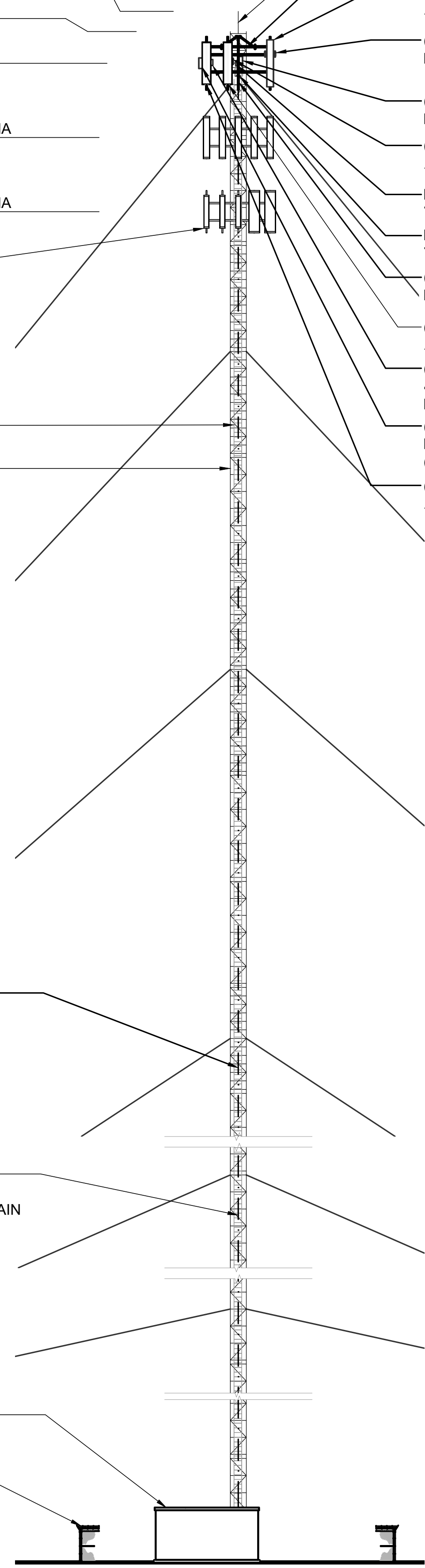
SHEET TITLE

EQUIPMENT PLAN

SHEET NUMBER

A2



NOTES:
THIS DRAWING IS FOR EXHIBIT AND LAYOUT PURPOSES ONLY.

PLEASE REFER TO STRUCTURAL DOCUMENTS (PREPARED BY OTHERS) FOR PROJECT STRUCTURAL CALCULATION AND RESULTS.

NO WORK IS TO BE DONE WITHOUT AN APPROVED STRUCTURAL ANALYSIS PROVIDED BY OTHERS.

SPECIAL NOTES:

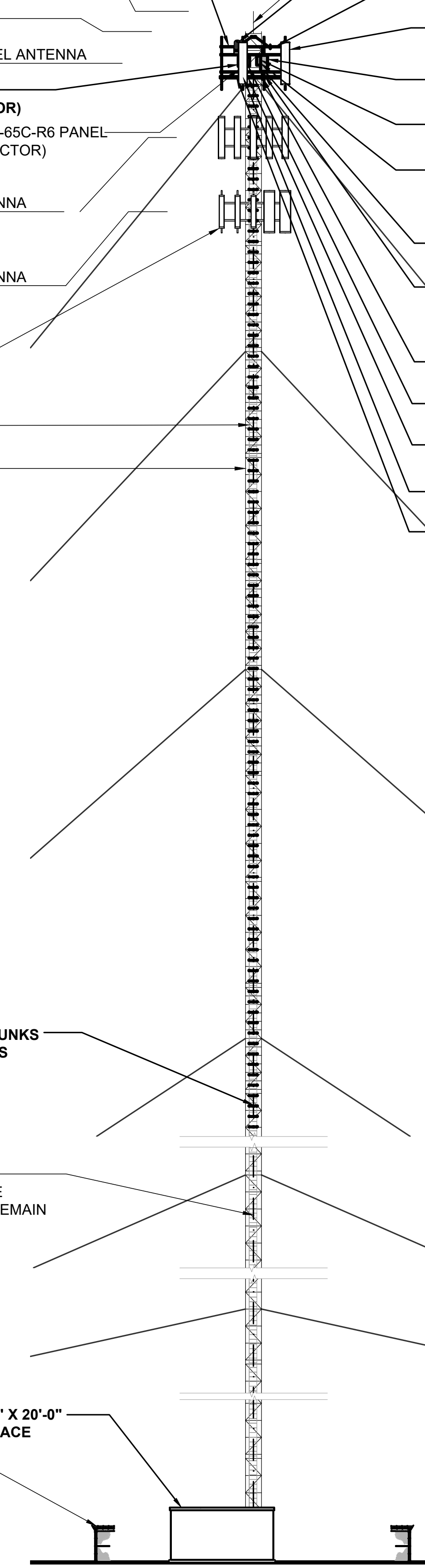
1. GC TO VERIFY ALL HEIGHTS AND AZIMUTHS IN FIELD PRIOR TO CONSTRUCTION. GC SHALL NOTIFY P.M. AND ARCHITECT/ENGINEER OF ANY DISCREPANCIES IMMEDIATELY.
2. STRUCTURAL/ DESIGN & ANALYSIS SHALL BE PERFORMED & APPROVED BY TOWER OWNER AND MANUFACTURER (STRUCTURAL ANALYSIS BY OTHERS)
3. STRUCTURAL ANALYSIS PERFORMED BY OTHERS. CONTRACTOR TO THOROUGHLY REVIEW THE TOWER STRUCTURAL ANALYSIS FOR INFORMATION PERTAINING TO TOWER UPGRADES, MOUNTING TYPES, ANTENNA HEIGHTS, AND CABLE ROUTING. ANY OTHER DISCREPANCIES BETWEEN THE DRAWINGS, STRUCTURAL ANALYSIS, AND TOWER PLANS SHOULD BE BROUGHT TO THE ATTENTION OF THE PROJECT MANAGER PRIOR TO BIDDING AND INSTALLATION.

EXISTING TOWER ELEVATION

16' 0" 4' 8' 12' 16'

22"x34" SCALE: 1/16" = 1'-0"
11"x17" SCALE: 1/32" = 1'-0"

1



PER MOUNT ANALYSIS COMPLETED BY AMERICAN TOWER CORPORATION, DATED 03/26/2025, THE EXISTING MOUNT MUST BE MODIFIED TO ADEQUATELY SUPPORT THE PROPOSED LOADING. THE MOUNT MODIFICATION DETAILED AT THE END OF THIS PLAN SET, MUST BE INSTALLED PRIOR TO THE INSTALLATION OF THE PROPOSED ANTENNAS AND OTHER EQUIPMENT.

PER STRUCTURAL ANALYSIS COMPLETED BY ATC, DATED 02/06/2025, THE EXISTING TOWER CAN ADEQUATELY SUPPORT THE PROPOSED LOADING.

PROPOSED TOWER ELEVATION

16' 0" 4' 8' 12' 16'

22"x34" SCALE: 1/16" = 1'-0"
11"x17" SCALE: 1/32" = 1'-0"

2

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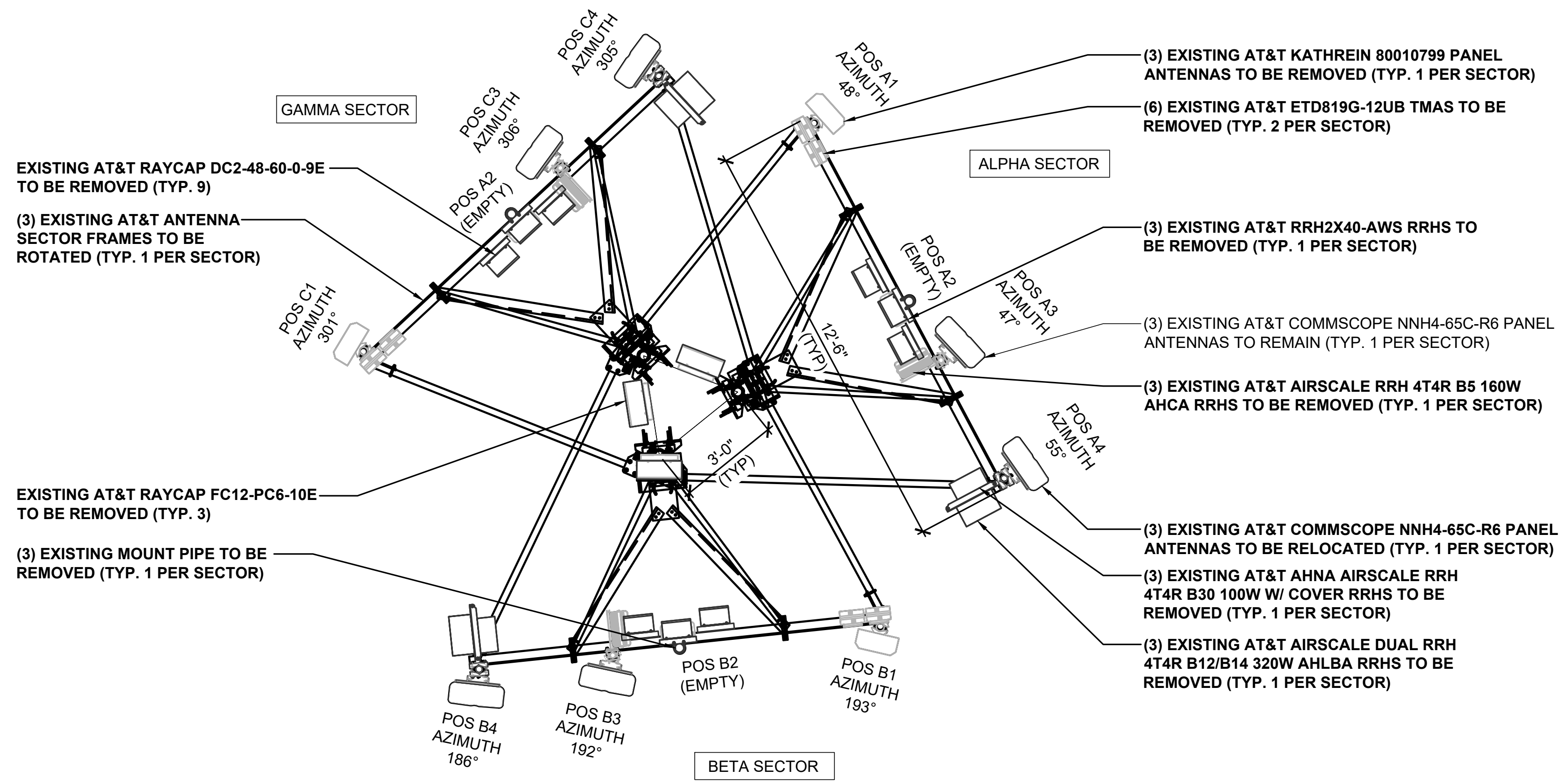
SHEET TITLE

TOWER ELEVATION

SHEET NUMBER

A3



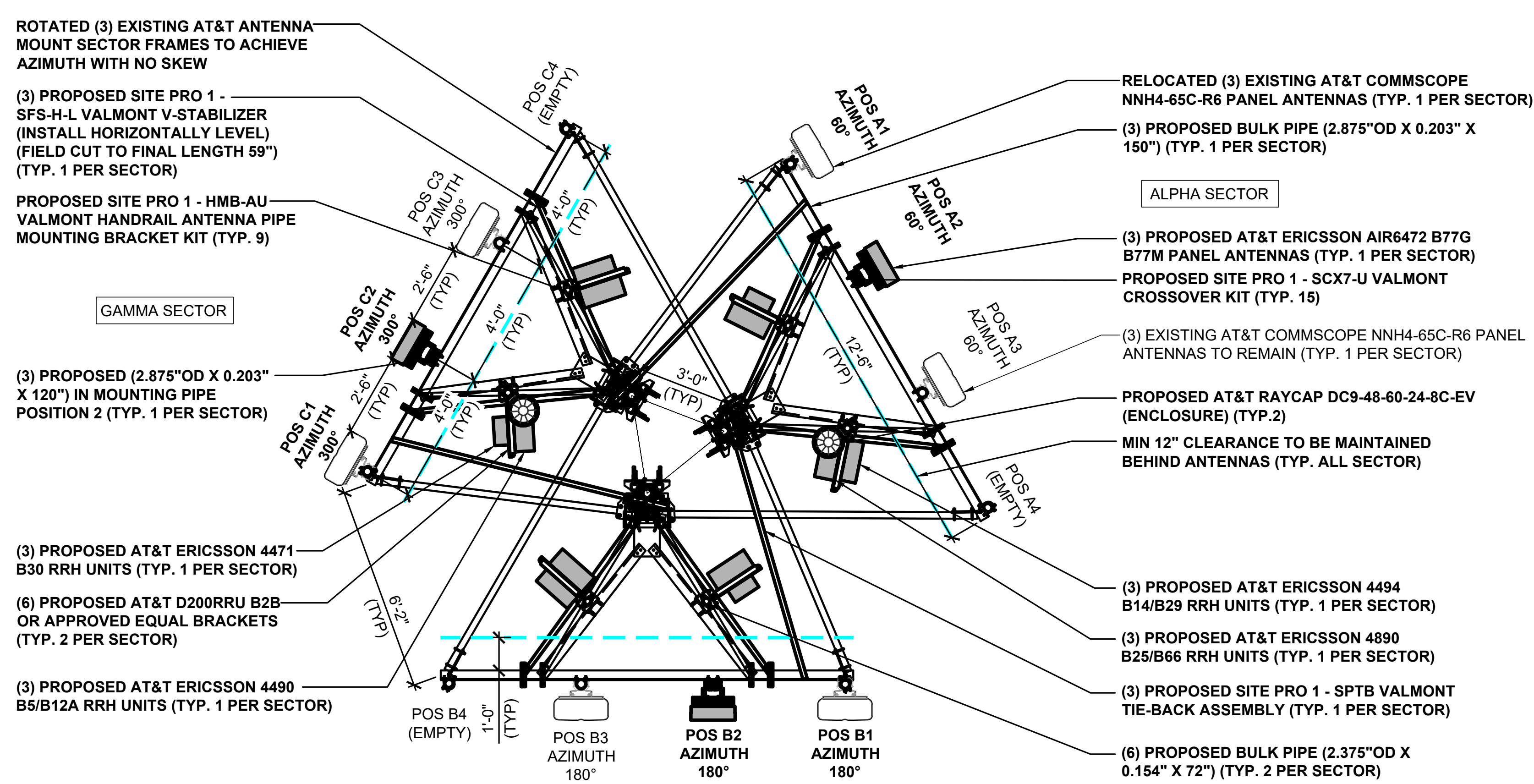
EXISTING ANTENNA PLAN

22"x34" SCALE: 3/8" = 1'-0"
11"x17" SCALE: 3/16" = 1'-0"

1

PER MOUNT ANALYSIS COMPLETED BY AMERICAN TOWER CORPORATION, DATED 03/26/2025, THE EXISTING MOUNT MUST BE MODIFIED TO ADEQUATELY SUPPORT THE PROPOSED LOADING. THE MOUNT MODIFICATION DETAILED AT THE END OF THIS PLAN SET, MUST BE INSTALLED PRIOR TO THE INSTALLATION OF THE PROPOSED ANTENNAS AND OTHER EQUIPMENT.

PER STRUCTURAL ANALYSIS COMPLETED BY ATC, DATED 02/06/2025, THE EXISTING TOWER CAN ADEQUATELY SUPPORT THE PROPOSED LOADING.



PROPOSED ANTENNA PLAN

22"x34" SCALE: 3/8" = 1'-0"
11"x17" SCALE: 3/16" = 1'-0"

2

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SHEET TITLE

ANTENNA PLAN

SHEET NUMBER

A4

PROPOSED ANTENNA CONFIGURATION AND CABLE SCHEDULE

SECTOR	POS	TECH	ANTENNA	ANTENNA ℄ HEIGHT	AZIMUTH	TMA/RRU MODEL #	DC SURGE AND DISTRIBUTION	CABLE TYPE	CABLE LENGTH (±20%)	DOWNTILTS
A	1	-	**COMMSCOPE NNH4-65C-R6 (XR)**	394' AGL	60°	**(1) 4490 B5/B12A (N) (1) 4471 B30 (N)**	**(2) RAYCAP (N) DC9-48-60-24-8C-EV (ENCLOSURE) (RAD = 394'-0")**	(6) COAX CABLES (X 1-5/8")	480'-0"	0
	2	**5G CBAND**	**ERICSSON AIR6472 B77G B77M (N)**			-		**(5) DC TRUNK LINES (N 6 GAUGE) (0.96") (2) FIBER LINES (N 24 PAIR) (0.40")**		0
	3	-	COMMSCOPE NNH4-65C-R6 (X)			**(1) 4494 B14/B29 (N) (1) 4890 B25/B66 (N)**		(1) RET LINE (X 3/8") **FIBER LINE SHARED W/A2**		0
	4	-	-			-		-		-
B	1	-	**COMMSCOPE NNH4-65C-R6 (XR)**	394' AGL	180°	**(1) 4490 B5/B12A (N) (1) 4471 B30 (N)**		DC TRUNK LINES SHARED W/A1 FIBER LINES SHARED W/A1	480'-0"	0
	2	**5G CBAND**	**ERICSSON AIR6472 B77G B77M (N)**			-		**DC TRUNK LINES SHARED W/A2 FIBER LINE SHARED W/A2**		0
	3	-	COMMSCOPE NNH4-65C-R6 (X)			**(1) 4494 B14/B29 (N) (1) 4890 B25/B66 (N)**		DC TRUNK LINE SHARED W/A3 **FIBER LINE SHARED W/A2**		0
	4	-	-			-		-		-
C	1	-	**COMMSCOPE NNH4-65C-R6 (XR)**	394' AGL	300°	**(1) 4490 B5/B12A (N) (1) 4471 B30 (N)**		DC TRUNK LINES SHARED W/A1 FIBER LINES SHARED W/A1	480'-0"	0
	2	**5G CBAND**	**ERICSSON AIR6472 B77G B77M (N)**			-		**DC TRUNK LINES SHARED W/A2 FIBER LINE SHARED W/A2**		0
	3	-	COMMSCOPE NNH4-65C-R6 (X)			**(1) 4494 B14/B29 (N) (1) 4890 B25/B66 (N)**		DC TRUNK LINE SHARED W/A3 **FIBER LINE SHARED W/A2**		0
	4	-	-			-		-		-

* INCLUDES SAFETY FACTOR OF 20' FT. (10 FT. AT BOTH ENDS OF CABLE RUN). CONTRACTOR TO VERIFY RF DATA AT&T WIRELESS CONSTRUCTION MANAGER AND/OR RF ENGINEER PRIOR TO INSTALLATION

(N) = NEW
(X) = EXISTING
(XR) = EXISTING/RELOCATED
(E) = ELECTRICAL
(M) = MECHANICAL

ANTENNA & CABLE CONFIGURATION | NOT TO SCALE | 1

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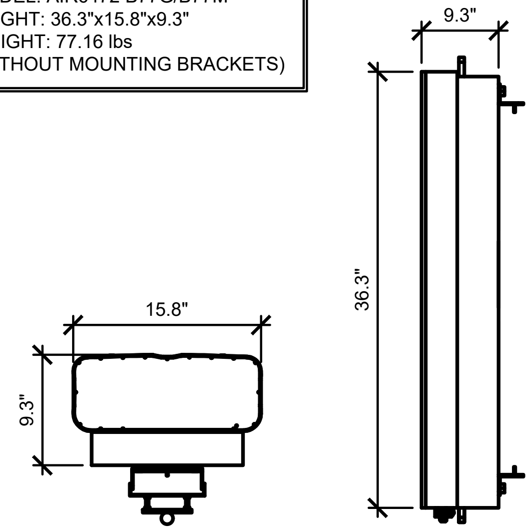
SHEET TITLE
ANTENNA & CABLE CONFIGURATION

SHEET NUMBER
A5

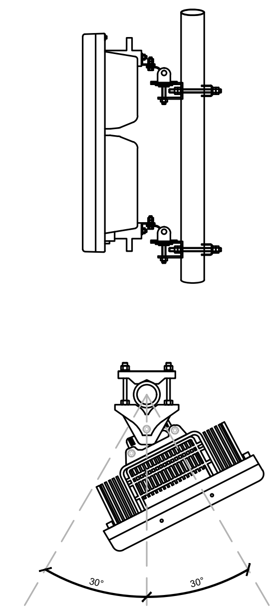
MANUFACTURER: ERICSSON
MODEL: AIR6472 B77G/B77M
HEIGHT: 36.3"x15.8"x9.3"
WEIGHT: 77.16 lbs
(WITHOUT MOUNTING BRACKETS)



TOP VIEW

SIDE VIEW

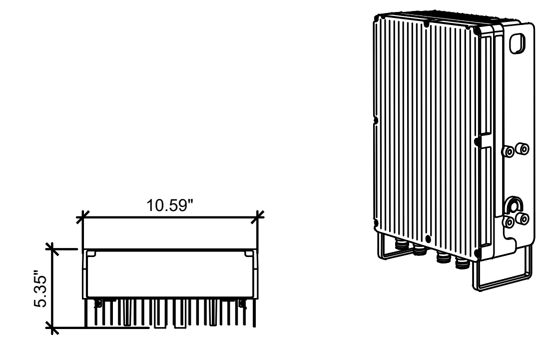
ANTENNA MANUFACTURER DETAIL - FOR REF. ONLY	NO SCALE	6



POLE	CIRCULER	SQURE	90° ANGLE
MINIMUM OUTER DIMENSION	Ø76mm	50X50mm	50X50mm
MAXIMUM OUTER DIMENSION	Ø114mm	80X80mm	80X80mm

ERICSSON - AIR ANTENNA MOUNTING DETAIL - FOR REF. ONLY	NO SCALE	4

MANUFACTURER: ERICSSON
MODEL: RADIO 4471 B30
DIMENSIONS, [HxWxD] IN: 16.1"x10.59"x5.35"
WEIGHT: 28.4 lbs

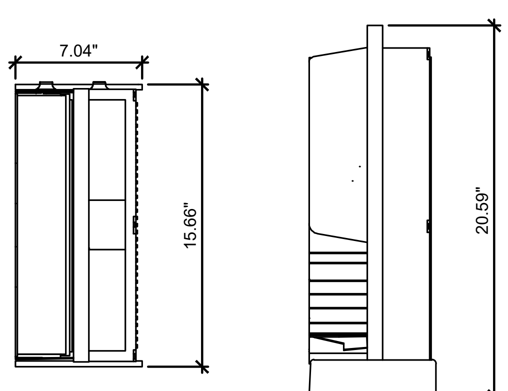


TOP VIEW

ISOMETRIC VIEW

RRH MANUFACTURER DETAIL - FOR REF. ONLY	NO SCALE	2

MANUFACTURER: ERICSSON
MODEL: RADIO 4890 B25/B66
DIMENSIONS, [HxWxD] IN: 20.59"x15.66"x7.04"
WEIGHT: 67.24 lbs

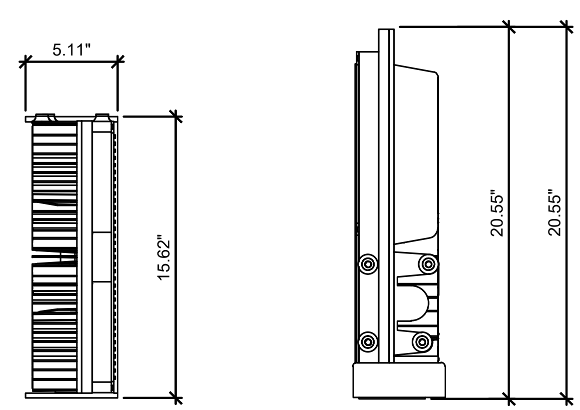


TOP VIEW

SIDE VIEW

RRH MANUFACTURER DETAIL - FOR REF. ONLY	NO SCALE	5

MANUFACTURER: ERICSSON
MODEL: RADIO 4490 B5/B12A
DIMENSIONS, [HxWxD] IN: 20.55"x15.62"x5.11"
WEIGHT: 50.26 lbs

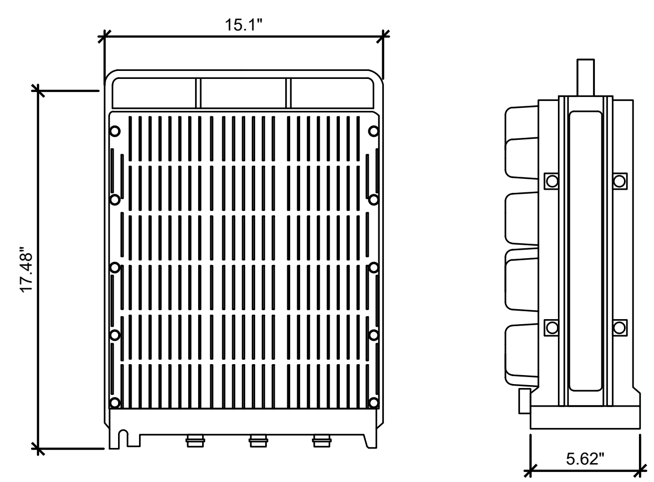


TOP VIEW

SIDE VIEW

RRH MANUFACTURER DETAIL - FOR REF. ONLY	NO SCALE	3

MANUFACTURER: ERICSSON
MODEL: RADIO 4494 B14/B29
DIMENSIONS, [HxWxD] IN: 17.48"X15.1"X5.62"
WEIGHT: 57 lbs



FRONT VIEW

SIDE VIEW

RRH DETAIL - FOR REFERENCE ONLY	NO SCALE	1

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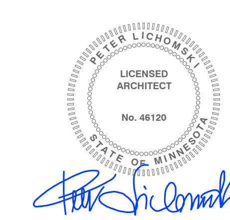

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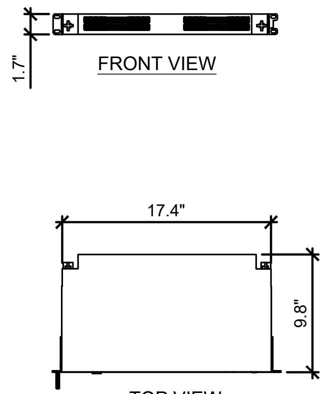
SHEET TITLE

ANTENNA, RRH AND MOUNTING DETAIL

SHEET NUMBER

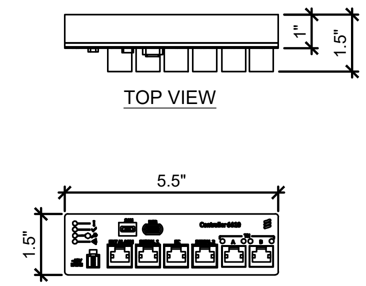
A6

MANUFACTURER: ERICSSON
MODEL: 6672
DIMENSIONS, [HxWxD] IN: 1.7"x17.4"x9.8"
WEIGHT: 13.2 lbs

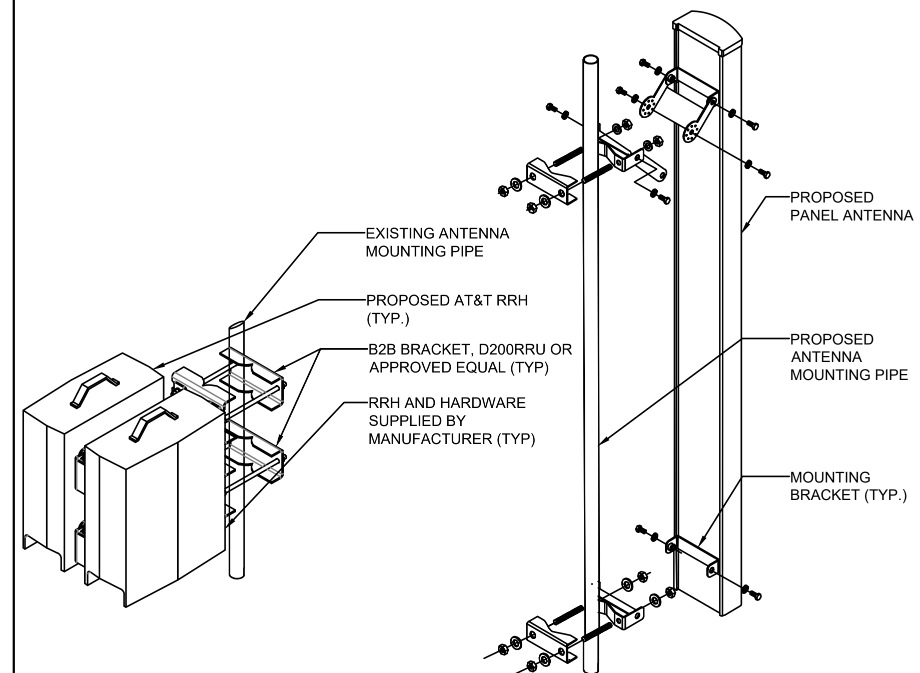


6672 MANUFACTURER DETAIL - FOR REF. ONLY | NO SCALE | 6

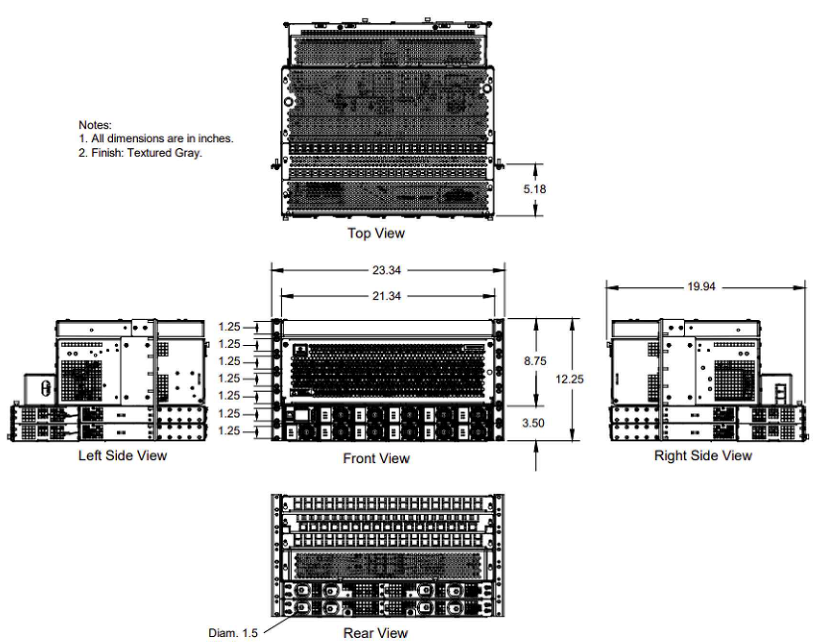
MANUFACTURER: ERICSSON
MODEL: 6610
DIMENSIONS, [HxWxD] IN: 1.57"x5.5""x1.0"
WEIGHT: 0.5 lbs



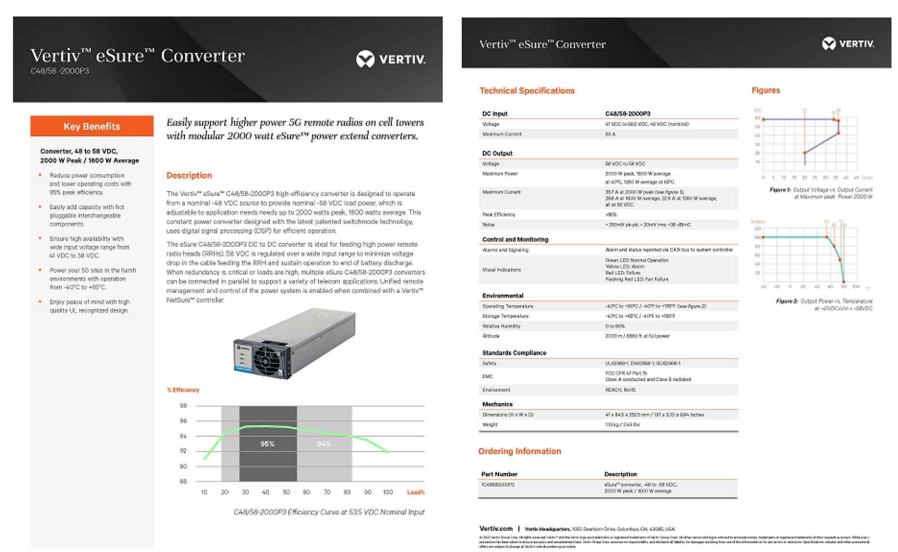
6601 MANUFACTURER DETAIL - FOR REF. ONLY | NO SCALE | 4



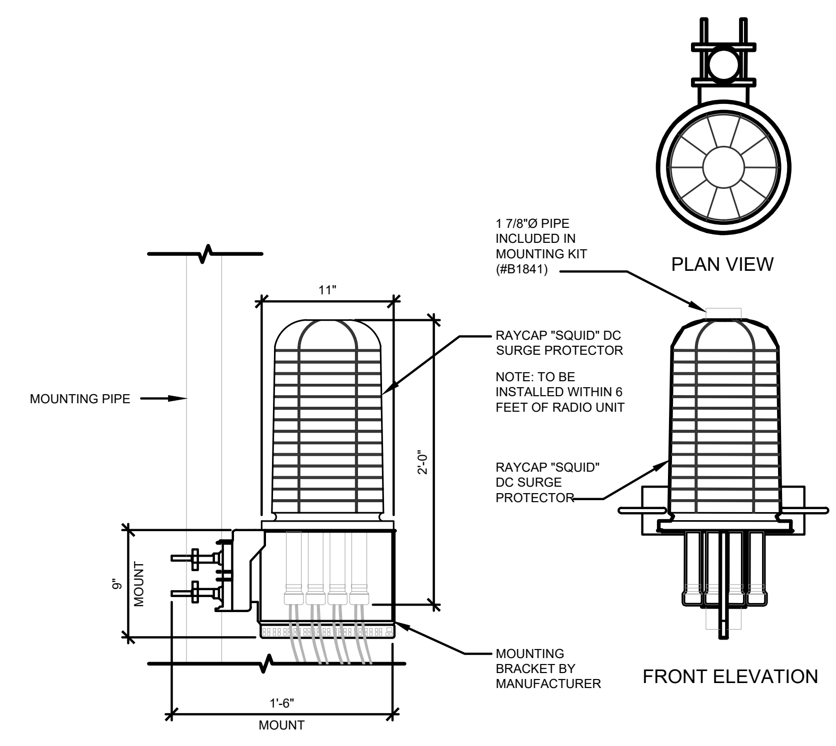
ANTENNA & RRH MOUNTING DETAIL - FOR REFERENCE ONLY | NO SCALE | 2



NETSURE 58V UPGRADE KIT DETAIL - FOR REF. ONLY | NO SCALE | 5



CONVERTER MANUFACTURER DETAIL - FOR REF. ONLY | NO SCALE | 3



MANUFACTURER: RAYCAP
MODEL: DC9-48-60-24-8C-EV "SQUID"

SQUID MANUFACTURER DETAIL - FOR REF. ONLY | NO SCALE | 1

AT&T
7900 XERXES AVE S,
3RD FLOOR
BLOOMINGTON, MN 55431

AMERICAN TOWER
SITE DESIGN
3500 REGENCY PARKWAY, STE 100
CARY, NORTH CAROLINA 27518
PHONE: (919) 468-0112

LAB
49030 Pontiac Trail, Suite 100
Wixom, Michigan 48393
PHONE: (248) 705-9212

REVISIONS

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PETER LICHOMSKI
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COOK, MN 55723-8420

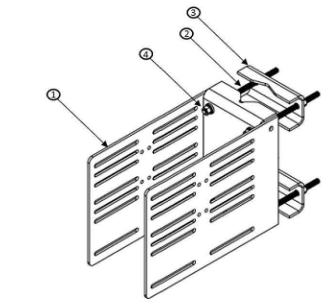
SHEET TITLE

ANTENNA, RRH AND MOUNTING DETAIL

SHEET NUMBER

A7

Technical Data Sheet	Rosenberger
LOW PIM 7 SLOTTED RRU MOUNT, UNIVERSAL	D200RRU



* Drawings not to scale

Description

LOW PIM 7 SLOTTED RRU MOUNT, UNIVERSAL

Parts

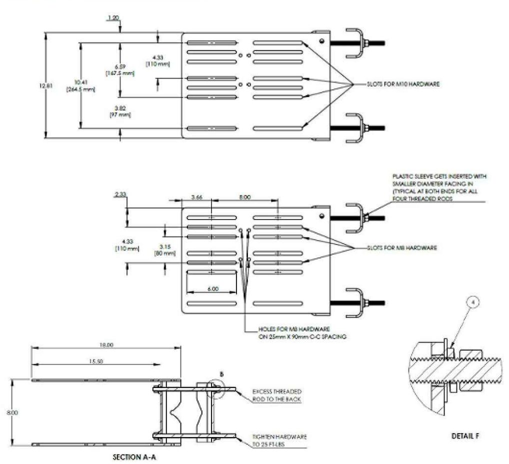
ITEM	QTY.	DESCRIPTION	WEIGHT [lb.]
1	1	RRU MOUNT WELDMENT,7 SLOTTED,LOW PIM	37.98
2	4	THREADED ROD ASSEMBLY 1/2" X 0'-10" (A36)	2.40
3	2	RRU MOUNT CLAMP (FORMED)	6.16
4	8	FLAT WASHER, LOCK 1/2" A153 GALV	0.08
		TOTAL WEIGHT	46.62

Technical Data Sheet	Rosenberger
LOW PIM 7 SLOTTED RRU MOUNT, UNIVERSAL	D200RRU

Notes

1. FITS ROUND LEGS: 1/2" TO 5"
60 DEG ANGLE LEGS 1" TO 5"
90 DEG ANGLE LEGS 1" TO 3 1/2"
2. THIS MOUNT IS STRUCTURALLY ADEQUATE TO SUPPORT (2) 38"H x 18"W x 11"D RADIOS (120LBS. EACH) CENTERED BETWEEN SLOTS AT AN ELEVATION OF 200' ABOVE GRADE, WITH STRUCTURE CLASS II, EXPOSURE CATEGORY C, TOPOGRAPHIC CATEGORY 1, AND A BASIC WIND SPEED OF 150 MPH (194 MPH ULTIMATE) PER ANSI/TIA-222-G.

Dimensions



GRIP ANCHOR, CABLE LADDER, 40" UNIVERSAL DETAIL - FOR REF. ONLY | NO SCALE | 4

860674608-U | GRIP ANCHOR,CABLE LADDER,40", UNIVERSAL

Heavy Duty Cable Ladder Cross Member with Cable Attachment Points

-Product is designed to be field installed as needed on cable ladders to allow for the attachment of cables. Replaces a current rung on the cable ladder. Additional units can be installed along the cable ladder as needed to support additional cables.

-Purpose built cable ladder rung that is designed to support 1000 pounds of cable per eye or up to 1000 pounds in combination. Ideal for Hybrid cable. Includes mounting eye bolt for attachment.

Installs on the below cable ladders up to 40" wide

-Not to be used for hoisting.
-Competent person to make determination of usage on the structure.
-Competent person to field cut to lenth as needed. Apply cold galv to cut ends.

Product Classification

Portfolio	CommScope®
Product Type	Ladder kit
Ordering Note	CommScope® standard product with terms
Warranty	One year

General Specifications

Cable Runs, quantity	3

Dimensions

Width	1,016 mm | 40 in

Material Specifications

Material Type	Hot dip galvanized steel

Packaging and Weights

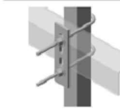
Included	Attachment kits
Packaging quantity	1

Included Products

HK-CL-U

COMMSCOPE®

HK-CL-U



Universal Cable Ladder Hardware Kit for 1 in to 4-1/2 in round or angle members

Product Classification

Portfolio	CommScope®
Product Type	Ladder kit
Ordering Note	CommScope® standard product (Global)

General Specifications

Mounting	Round and angle legs

Dimensions

Height	203.2 mm | 8 in
Width	127 mm | 5 in
Length	203.2 mm | 8 in
Mounting Diameter, maximum	114.3 mm | 4.5 in
Mounting Diameter, minimum	25.4 mm | 1 in

Material Specifications

Material Type	Hot dip galvanized steel

Packaging and Weights

Included	J-bolts | Plates
Packaging quantity	1
Weight, net	1.7 kg | 3.748 lb

Regulatory Compliance/Certifications

Agency	Classification
CHINA-ROHS	Below maximum concentration value
ISO 9001:2015	Designed, manufactured and/or distributed under this quality management system
REACH-SVHC	Compliant as per SVHC revision on www.commscope.com/ProductCompliance

COMMSCOPE®

GRIP ANCHOR, CABLE LADDER, 40" UNIVERSAL DETAILS - FOR REF. ONLY | NO SCALE | 2

Pxxx: Bulk Pipe

SITE PRO 1 — A valmont company

Part #	Length	OD x Length (in)
Schedule 40		
P260	5'-0"	2-3/8" x 60"
P263	5'-3"	2-3/8" x 63"
P272	6'-0"	2-3/8" x 72"
P284	7'-0"	2-3/8" x 84"
P296	8'-0"	2-3/8" x 96"
P2108	9'-0"	2-3/8" x 108"
P2120	10'-0"	2-3/8" x 120"
P2126	10'-6"	2-3/8" x 126"
P2150	12'-6"	2-3/8" x 150"
P2174	14'-6"	2-3/8" x 174"
P2252	21'-0"	2-3/8" x 252"
P3072	6'-0"	2-7/8" x 72"
P3084	7'-0"	2-7/8" x 84"
P3096	8'-0"	2-7/8" x 96"
P30108	9'-0"	2-7/8" x 108"
P30120	10'-0"	2-7/8" x 120"
P30126	10'-6"	2-7/8" x 126"
P30150	12'-6"	2-7/8" x 150"
P30174	14'-6"	2-7/8" x 174"
P30252	21'-0"	2-7/8" x 252"
P360	5'-0"	3-1/2" x 60"
P372	6'-0"	3-1/2" x 72"
P384	7'-0"	3-1/2" x 84"
P396	8'-0"	3-1/2" x 96"
P3150	12'-6"	3-1/2" x 150"
P3160	13'-4"	3-1/2" x 160"
P3174	14'-6"	3-1/2" x 174"
P3216	18'-0"	3-1/2" x 216"
P3252	21'-0"	3-1/2" x 252"
P472	6'-0"	4-1/2" x 72"
P4126	10'-6"	4-1/2" x 126"
P4252	21'-0"	4-1/2" x 252"

Features:
- Factory cut end, hot-dip galvanized pipe

Construction:
- ASTM A53 Grade B
- Schedule 40 or Schedule 80

Design Criteria:
- ASTM A53 Grade B (Yield Fy = 35 ksi [240 MPa]/ Tensile Fu = 60 ksi [415 MPa])
- Hot dip galvanized in accordance with ASTM A123 requirements

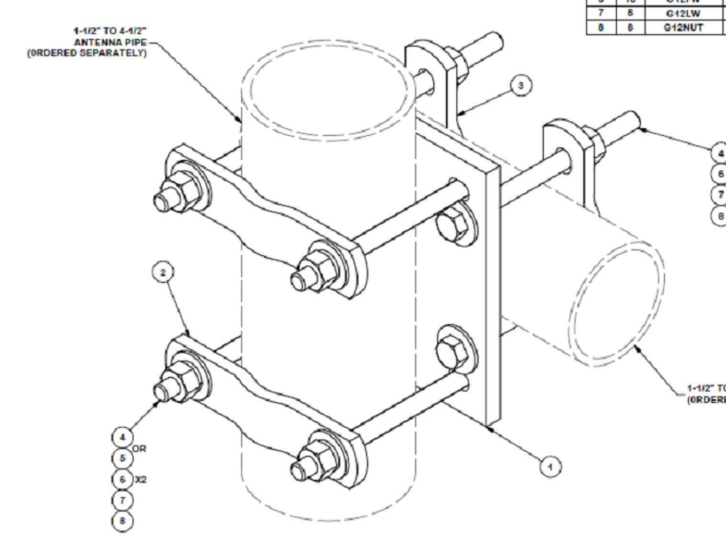
Part #	Length	OD x Length (in)
Schedule 80		
P2252-80	21'	2-1/2" x 252"
P30126-80	10'-6"	2-7/8" x 126"
P30252-80	21'	2-7/8" x 252"
P3252-80	21'	3-1/2" x 252"

created on: 03/09/2023

SitePro1.com 888-438-7761

MOUNT PIPES DETAIL - FOR REF. ONLY | NO SCALE | 3

PARTS LIST						
ITEM	QTY	PART NO.	PART DESCRIPTION	LENGTH	UNIT WT.	NET WT.
1	1	SCX7	CROSSOVER PLATE	8 in	7.55	7.55
2	2	X-115765	5" V-CLAMP		1.02	2.04
3	2	X-100064	CLAMP (S) (4" V-CLAMP) GALVANIZED		0.91	1.83
4	8	G12065	1/2" x 6-1/2" HDG HEX BOLT GR5 FULL THREAD	6 1/2 in	0.41	3.26
5	4	G12045	1/2" x 4.5" HDG HEX BOLT GR5 FULL THREAD	4 1/2 in	0.30	1.19
6	16	G12FW	1/2" HDG USS FLATWASHER		0.03	0.54
7	8	G12LW	1/2" HDG LOCKWASHER		0.01	0.11
8	8	G12NUT	1/2" HDG HEAVY 2H HEX NUT		0.07	0.57
					TOTAL WT. #	16.98



TOLERANCE NOTES

TOLERANCES ON DIMENSIONS, UNLESS OTHERWISE NOTED ARE:
SAWED, SHEARED AND GAS CUT EDGES (± 0.030")
DRILLED AND GAS CUT HOLES (± 0.030") - NO CONING OF HOLES
LASER CUT EDGES AND HOLES (± 0.010") - NO CONING OF HOLES
BENDS ARE ± 1/2 DEGREE
ALL OTHER MACHINING (± 0.030")
ALL OTHER ASSEMBLY (± 0.060")

PROPRIETARY NOTE: THE DATA AND TECHNIQUES CONTAINED IN THIS DRAWING ARE PROPRIETARY INFORMATION OF VALMONT INDUSTRIES AND CONSIDERED A TRADE SECRET. ANY USE OR DISCLOSURE WITHOUT THE CONSENT OF VALMONT INDUSTRIES IS STRICTLY PROHIBITED.

DESCRIPTION: CROSSOVER PLATE (V-CLAMP STYLE)

SITE PRO 1 — A valmont company — Engineering Support Team: 1-888-753-7446 — Locations: New York, NY; Atlanta, GA; Los Angeles, CA; Plymouth, IN; Salem, OR; Dallas, TX

CPD NO.	DRAWN BY	ENG. APPROVAL	PART NO.	PAGE
	CEK 10/7/2010		SCX7-U	1 OF 1
CLASS 81 / SUB 01	DRAWING USAGE: CUSTOMER	CHECKED BY: BMC 10/8/2010	DWG. NO.: SCX7-U	

SCX7-U CROSSOVER PLATE DETAIL - FOR REF. ONLY | NO SCALE | 1

AT&T
7900 XERXES AVE S, 3RD FLOOR
BLOOMINGTON, MN 55431

AMERICAN TOWER
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3500 REGENCY PARKWAY, STE 100
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LAB
49030 Pontiac Trail, Suite 100
Wixom, Michigan 48393
PHONE: (248) 705-9212

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PETER LICHOMSKI — LICENSED ARCHITECT — No. 46120 — STATE OF MINNESOTA

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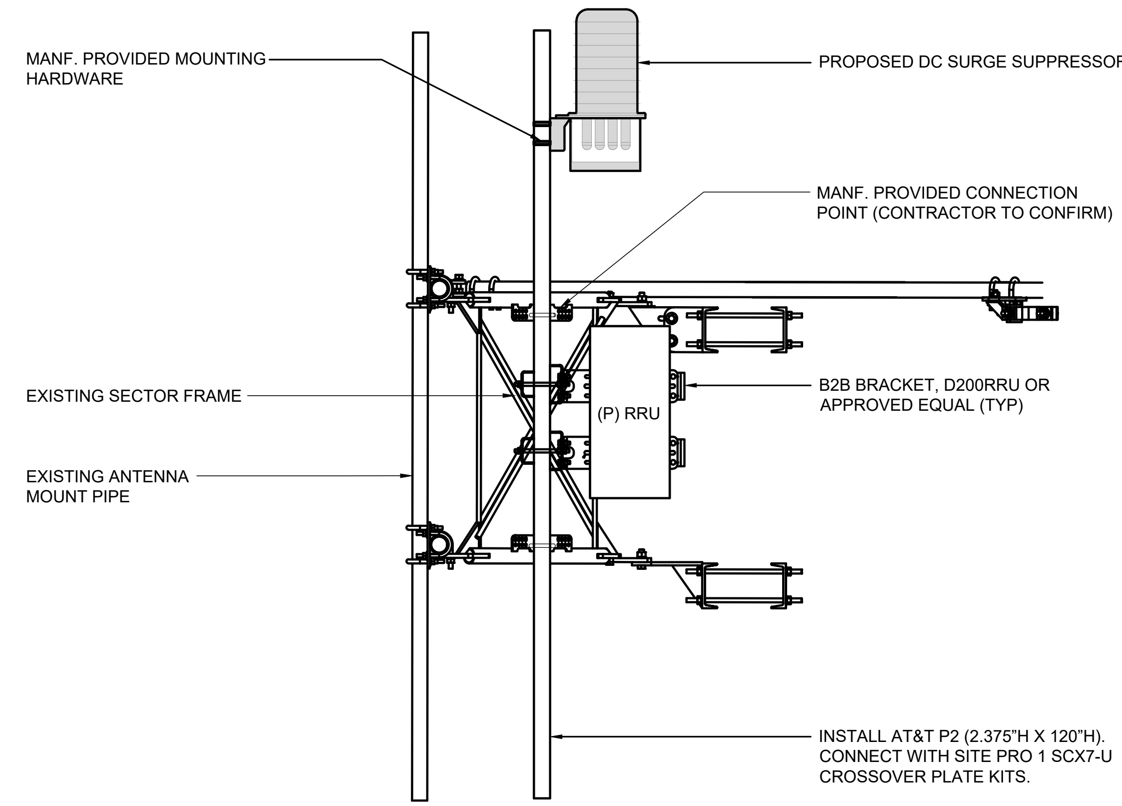
7393 FRAZIER ROAD,
COOK, MN 55723-8420

SHEET TITLE

ANTENNA, RRH AND MOUNTING DETAIL

SHEET NUMBER

A7A



NOT USED | NO SCALE | 4

RRU & DC SURGE MOUNTING DETAIL - FOR REF. ONLY | NO SCALE | 2

NOT USED | NO SCALE | 3

NOT USED | NO SCALE | 1

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SHEET TITLE

ANTENNA, RRH AND MOUNTING DETAIL

SHEET NUMBER

A7B

NOT USED | NO SCALE | 6

NOT USED | NO SCALE | 5

1. CONTRACTOR IS TO REFER TO AT&T'S MOST CURRENT RADIO FREQUENCY DATA SHEET (RFDS) PRIOR TO CONSTRUCTION.
2. THE SIZE, HEIGHT AND DIRECTION OF THE ANTENNAS SHALL BE ADJUSTED TO ACHIEVE THE AZIMUTHS SPECIFIED AND LIMIT SHADOWING AND TO MEET THE SYSTEM REQUIREMENTS.
3. CONTRACTOR SHALL VERIFY THE HEIGHT OF THE ANTENNA WITH THE AT&T WIRELESS PROJECT MANAGER.
4. VERIFY TYPE AND SIZE OF TOWER LEG PRIOR TO ORDERING ANY ANTENNA MOUNT.
5. UNLESS NOTED OTHERWISE THE CONTRACTOR MUST PROVIDE ALL MATERIAL NECESSARY.
6. ANTENNA AZIMUTHS ARE DEGREES OFF OF TRUE NORTH, BEARING CLOCKWISE, IN WHICH ANTENNA FACE IS DIRECTED. ALL ANTENNAS (AND SUPPORTING STRUCTURES AS PRACTICAL) SHALL BE ACCURATELY ORIENTED IN THE SPECIFIED DIRECTION.
7. CONTRACTOR SHALL VERIFY ALL RF INFORMATION PRIOR TO CONSTRUCTION.
8. SWEEP TEST SHALL BE PERFORMED BY GENERAL CONTRACTOR AND SUBMITTED TO AT&T WIRELESS CONSTRUCTION SPECIALIST. TEST SHALL BE PERFORMED PER AT&T WIRELESS STANDARDS.
9. CABLE LENGTHS WERE DETERMINED BASED ON THE DESIGN DRAWING. CONTRACTOR TO VERIFY ACTUAL LENGTH DURING PRE-CONSTRUCTION WALK.
10. CONTRACTOR TO USE ROSENBERGER FIBER LINE HANGER COMPONENTS (OR ENGINEER APPROVED EQUAL).

ANTENNA & CABLING NOTES | NO SCALE | 4

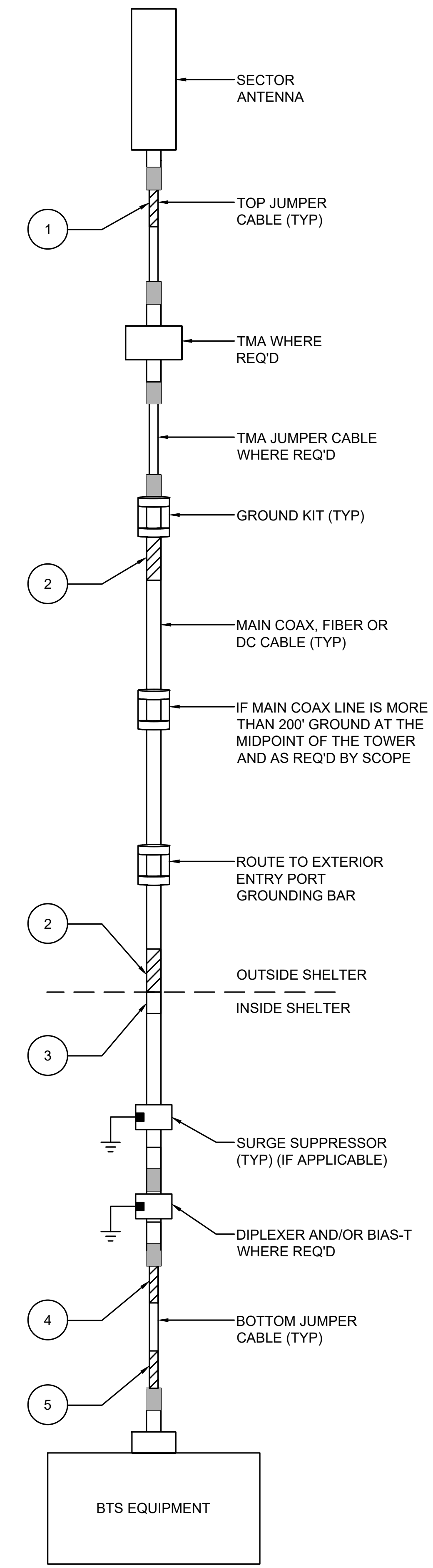
RF, DC & COAX CABLE MARKING LOCATIONS TABLE

NO	LOCATIONS
1	EACH TOP-JUMPER SHALL BE COLOR CODED WITH (1) SET OF 3" WIDE BANDS.
2	EACH MAIN COAX SHALL BE COLOR CODED WITH (1) SET OF 3" WIDE BANDS NEAR THE TOP-JUMPER CONNECTION AND WITH (1) SET OF 3/4" WIDE COLOR BANDS JUST PRIOR TO ENTERING THE BTS OR TRANSMITTER BUILDING.
3	CABLE ENTRY PORT ON THE INTERIOR OF THE SHELTER.
4	ALL BOTTOM JUMPERS SHALL BE COLOR CODED WITH (1) SET OF 3/4" WIDE BANDS ON EACH END OF THE BOTTOM JUMPER.
5	ALL BOTTOM JUMPERS SHALL BE COLOR CODED WITH (1) SET OF 3/4" WIDE BANDS ON EACH END OF THE BOTTOM JUMPER.

CABLE MARKING DIAGRAM | NO SCALE | 3

1. THE ANTENNA SYSTEM COAX SHALL BE LABELED WITH VINYL TAPE.
2. THE STANDARD IS BASED ON EIGHT COLORED TAPES-RED, BLUE, GREEN, YELLOW, ORANGE, BROWN, WHITE AND VIOLET. THESE TAPES MUST BE 3/4" WIDE & UV RESISTANT SUCH AS SCOTCH 35 VINYL ELECTRICAL COLOR CODING TAPE AND SHOULD BE READILY AVAILABLE TO THE ELECTRICIAN OR CONTRACTOR ON SITE.
3. USING COLOR BANDS ON THE CABLES, MARK ALL RF CABLE BY SECTOR AND CABLE NUMBER AS SHOWN ON "CABLE COLOR CHART".
4. WHEN AN EXISTING COAXIAL LINE THAT IS INTENDED TO BE A SHARED LINE BETWEEN TECHNOLOGIES IS ENCOUNTERED, THE CONTRACTOR SHALL REMOVE THE EXISTING COLOR CODING SCHEME AND REPLACE IT WITH THE COLOR CODING STANDARD. IN THE ABSENCE OF AN EXISTING COLOR CODING AND TAGGING SCHEME, OR WHEN INSTALLING PROPOSED COAXIAL CABLES, THIS GUIDELINE SHALL BE IMPLEMENTED AT THAT SITE REGARDLESS OF TECHNOLOGY.
5. ALL COLOR CODE TAPE SHALL BE 3M-35 AND SHALL BE INSTALLED USING A MINIMUM OF (3) THREE WRAPS OF TAPE AND SHALL BE NEATLY TRIMMED AND SMOOTHED OUT SO AS TO AVOID UNRAVELING.
6. ALL COLOR BANDS INSTALLED AT THE TOP OF THE TOWER SHALL BE A MINIMUM OF 3" WIDE AND SHALL HAVE A MINIMUM OF 3/4" OF SPACE BETWEEN EACH COLOR.
7. ALL COLOR CODES SHALL BE INSTALLED SO AS TO ALIGN NEATLY WITH ONE ANOTHER FROM SIDE-TO-SIDE.
8. IF EXISTING CABLES AT THE SITE ALREADY HAVE A COLOR CODING SCHEME AND THEY ARE NOT INTENDED TO BE REUSED OR SHARED WITH THE NEW TECHNOLOGY, THE EXISTING COLOR CODING SCHEME SHALL REMAIN UNTOUCHED.

CABLE MARKING DIAGRAM | NO SCALE | 2



CABLE COLOR CODING DIAGRAM | NO SCALE | 1

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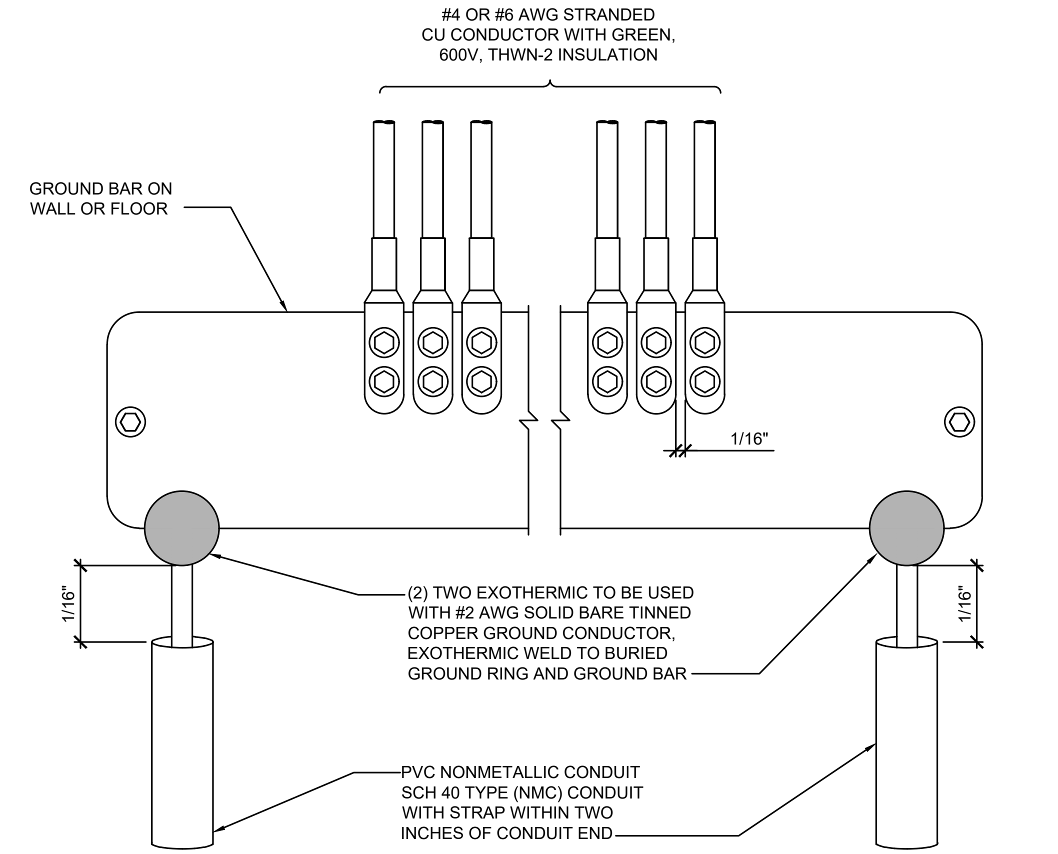
SHEET TITLE

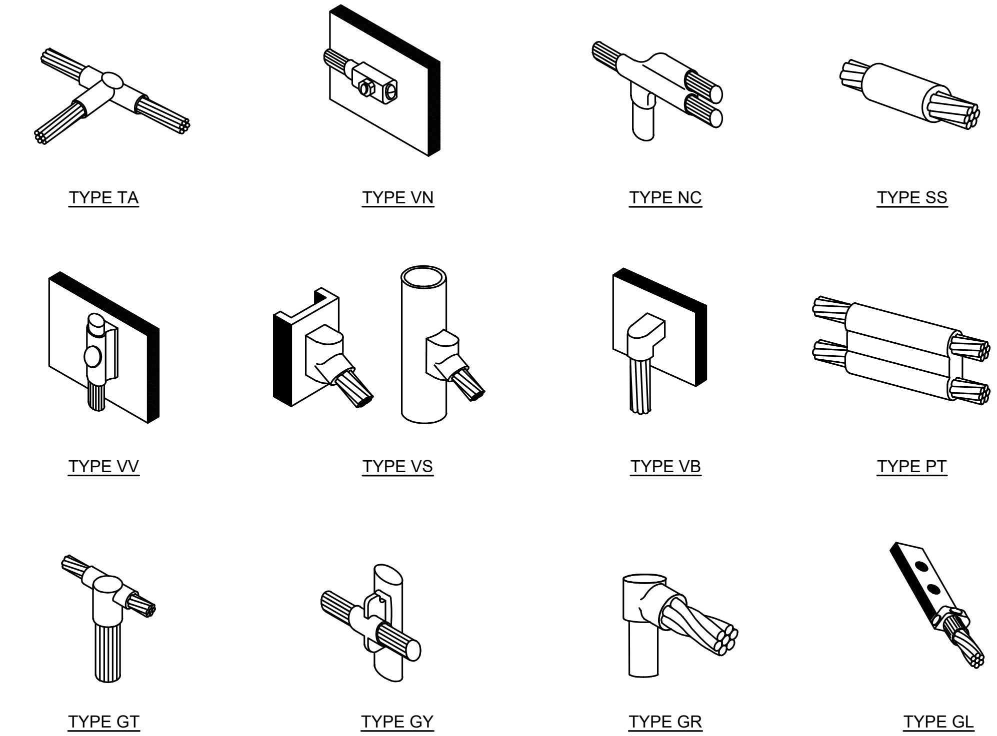
CABLE NOTES & COLOR CODING

SHEET NUMBER

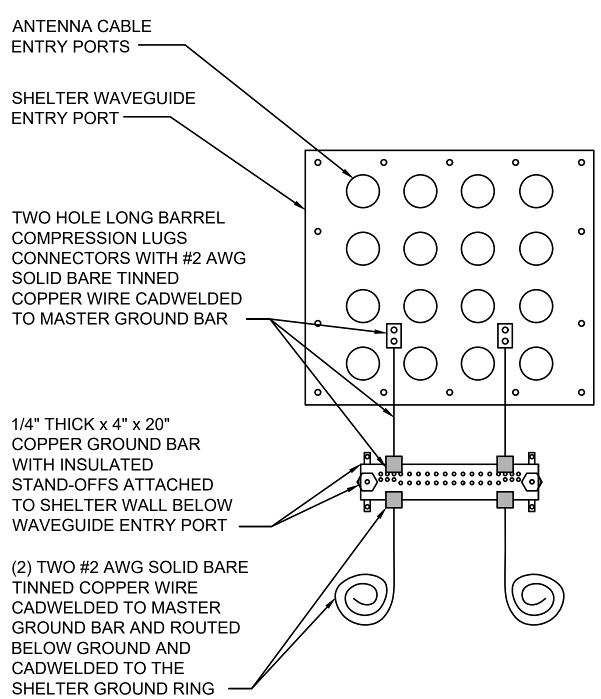
A8



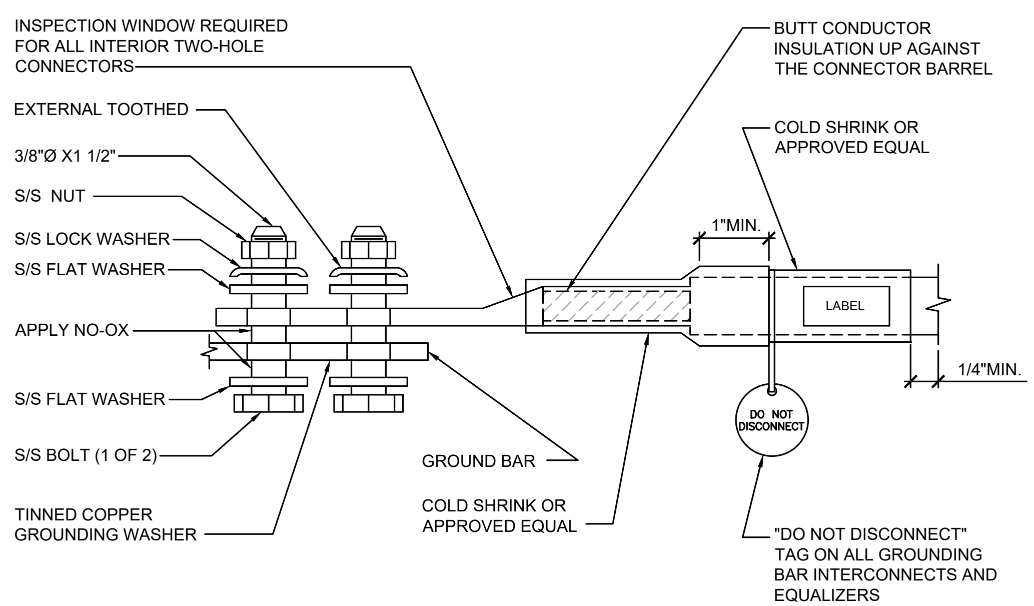
GROUND BAR DETAILS - FOR REFERENCE ONLY | NO SCALE | 5



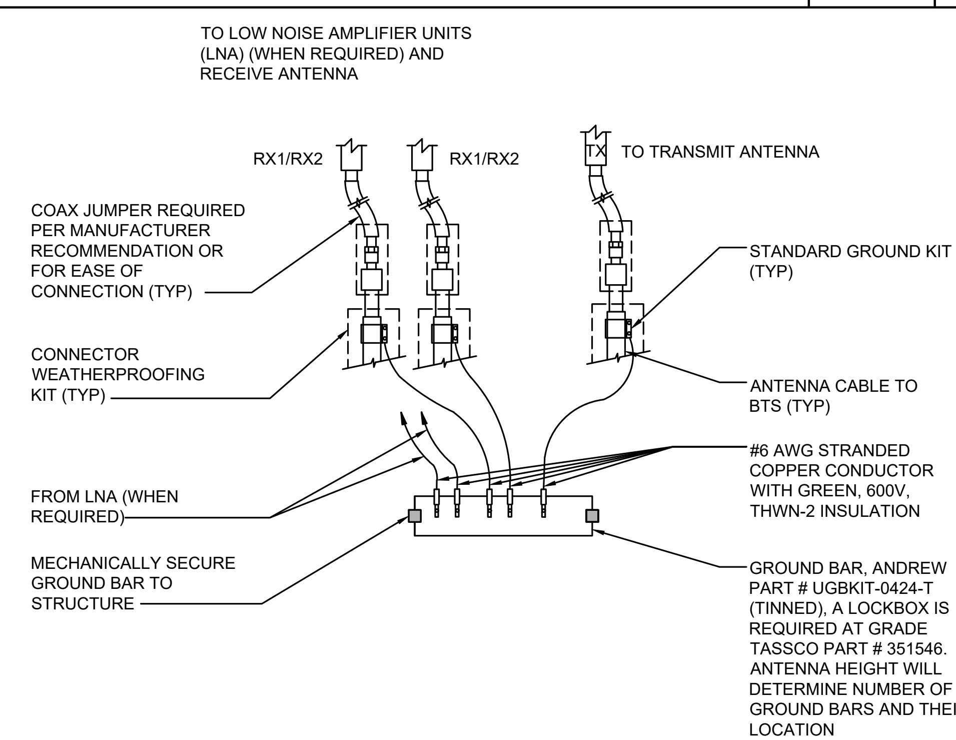
CADWELD DETAILS - FOR REFERENCE ONLY | NO SCALE | 4



COAX GROUND KIT DETAIL - FOR REFERENCE ONLY | NO SCALE | 3



EXTERIOR TWO HOLE LUG DETAIL - FOR REFERENCE ONLY | NO SCALE | 2



ANTENNA GROUND BAR DETAIL - FOR REFERENCE ONLY | NO SCALE | 1

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7900 XERXES AVE S,
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7393 FRAZIER ROAD,
COOK, MN 55723-8420

SHEET TITLE

GROUNDING DETAILS

SHEET NUMBER

E1

AMERICAN TOWER®
CORPORATION

Post Modification Mount Analysis Report

Mount Type	:	12.5 ft V-Frame
ATC Asset Name	:	COOK MN
ATC Asset Number	:	311180
Engineering Number	:	14865380_C9_04
Mount Elevation	:	393.5 ft
Proposed Carrier	:	AT&T Mobility
Carrier Site Name	:	LAKE VERMILLION
Carrier Site Number	:	WSUMW0041940
Site Location	:	7393 Frazier Road Cook, MN 55723-8420 47.8604, -92.4848
County	:	St. Louis
Date	:	March 26, 2025
Max Usage	:	88%
Analysis Result	:	Contingent Pass

Prepared By:
Max Carter
Structural Engineer II
Max Carter

I HEREBY CERTIFY THAT THIS PLAN, SPECIFICATION, OR REPORT WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA.
SIGNATURE:
NAME: SANTHOSHA SHANBHOGUE
LICENSE NUMBER: 56708
DATE:

Digitally Signed: 2025-03-28

Introduction

The purpose of this report is to summarize results of the mount analysis performed for AT&T Mobility at 393.5 ft.

Supporting Documents

Specifications Sheet:	Commscope SF-SU, dated March 20, 2014
Mount Modification:	TEP Project #31072.270238, dated July 3, 2019
Previous Analysis:	TEP Project #31072.258972, dated June 16, 2019
Radio Frequency Data Sheet:	RFDS ID #10129162, dated October 16, 2024
Reference Photos:	Site photos from 2021

Analysis

This mount was analyzed using American Tower Corporation's Mount Analysis Program and RISA-3D

Basic Wind Speed:	106 mph (3-Second Gust)
Basic Wind Speed w/ Ice:	50 mph (3-Second Gust) w/ 1.00" radial ice concurrent
Codes:	ANSI/TIA-222-I
Exposure Category:	D
Risk Category:	II
Topographic Factor Procedure:	Method 1
Feature:	Flat
Crest Height (H):	0 ft
Crest Length (L):	0 ft
Spectral Response:	Sds = 0.038, Sd1 = 0.016
Site Class:	D - Stiff Soil
Live Loads:	Lm = 500 lbs, Lv = 250 lbs

*Live Load(s) reduction is confirmed to either not govern or not be applicable

Conclusion

Based on the analysis results, the antenna mount meets the requirements per the applicable codes listed above provided the modifications listed below are completed:

- Install modification per ATC Drawing #14865380_C9_04

If you have any questions or require additional information, please reach out to your American Tower contact. If you do not have an American Tower contact and have an Engineering question, please contact MountAnalysis@americantower.com. Please include the American Tower site name, site number, and engineering number in the subject line for any questions.

AT&T
7900 XERXES AVE S,
3RD FLOOR
BLOOMINGTON, MN 55431

AMERICAN TOWER
SITE DESIGN
3500 REGENCY PARKWAY, STE 100
CARY, NORTH CAROLINA 27518
PHONE: (919) 468-0112

E/// MODERNIZATION
10129162
LAKE VERMILLION
7393 FRAZIER ROAD,
COOK, MN 55723-8420

SHEET TITLE
SUPPLEMENTAL

SHEET NUMBER
R1

Application Loading

Mount Centerline (ft)	Equipment Centerline (ft)	Qty	Equipment Manufacturer & Model
393.5	394.0	3	Ericsson AIR 6472 B77G B77M (92.6lbs)
		6	Commscope NNH4-65C-R6
		2	Raycap DC9-48-60-24-8C-EV (Enclosure)
		3	Ericsson Radio 4471 B30 (28.4lbs)
		3	Ericsson Radio 4490HP 44B5 44B12A C (20.6" Height)
		3	Ericsson Radio 4494 44B14 20B29 M01
		3	Ericsson Radio 4890HP 48B2/B25 48B66 M01 (68.3lbs)

Structure Usages

Structural Component	Controlling Usage	Pass/Fail
Horizontals	88%	Pass
Diagonals	21%	Pass
Tie-Backs	9%	Pass
Mount Pipes	25%	Pass
Tower Leg Check	59%	Pass
Clamp Connection Check	64%	Pass

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SHEET NUMBER
R2